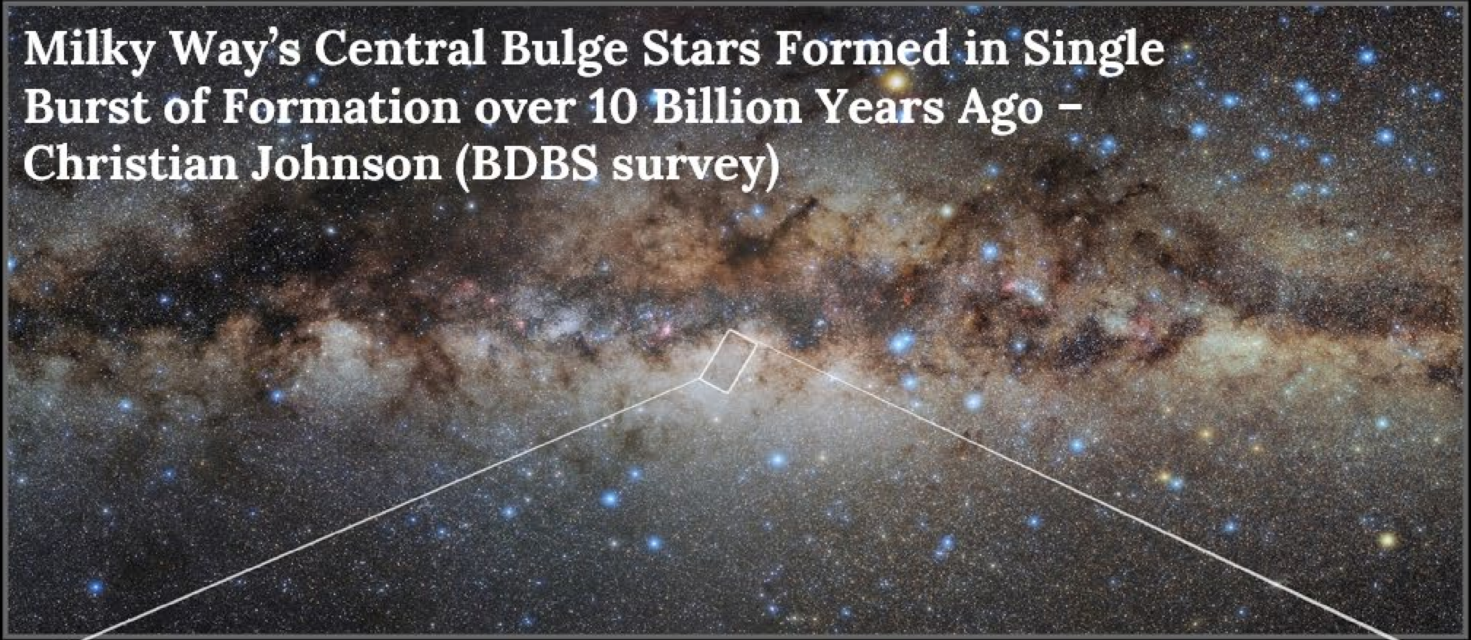
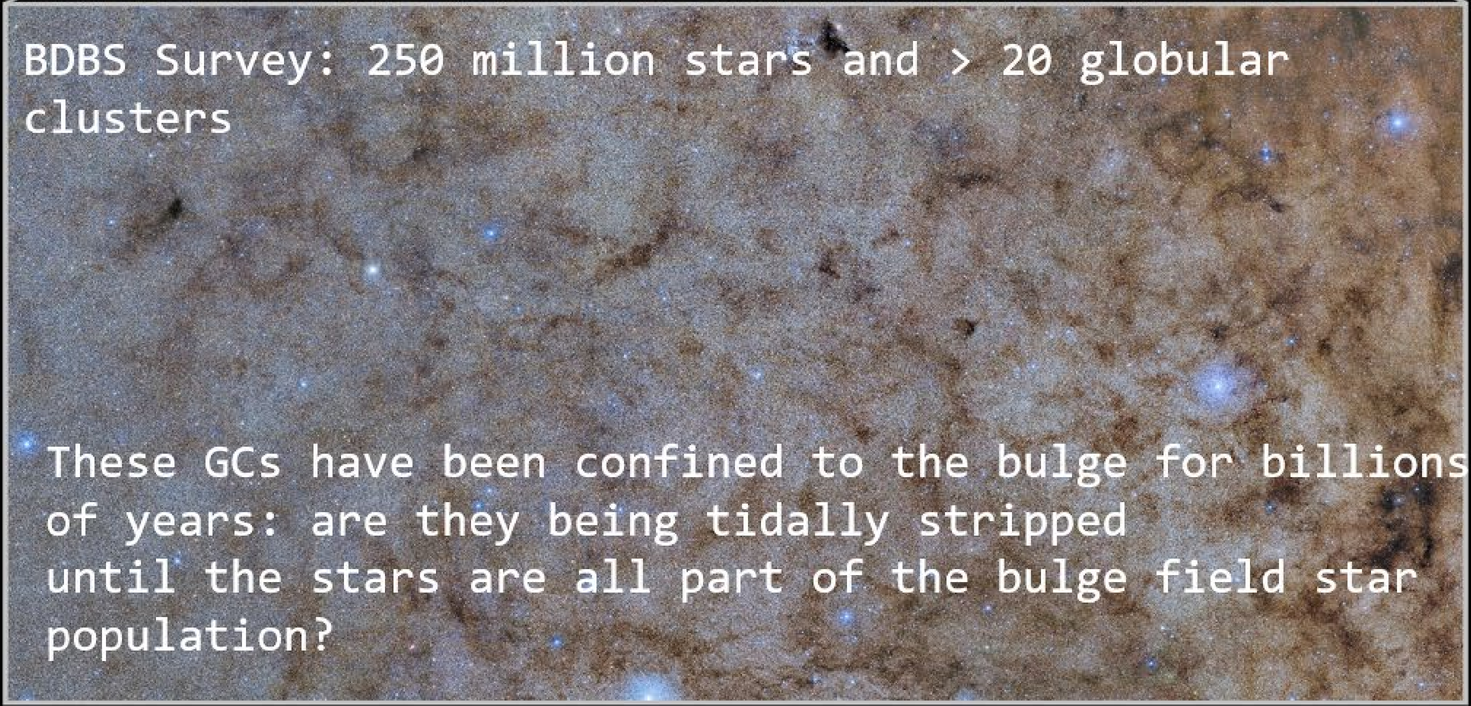
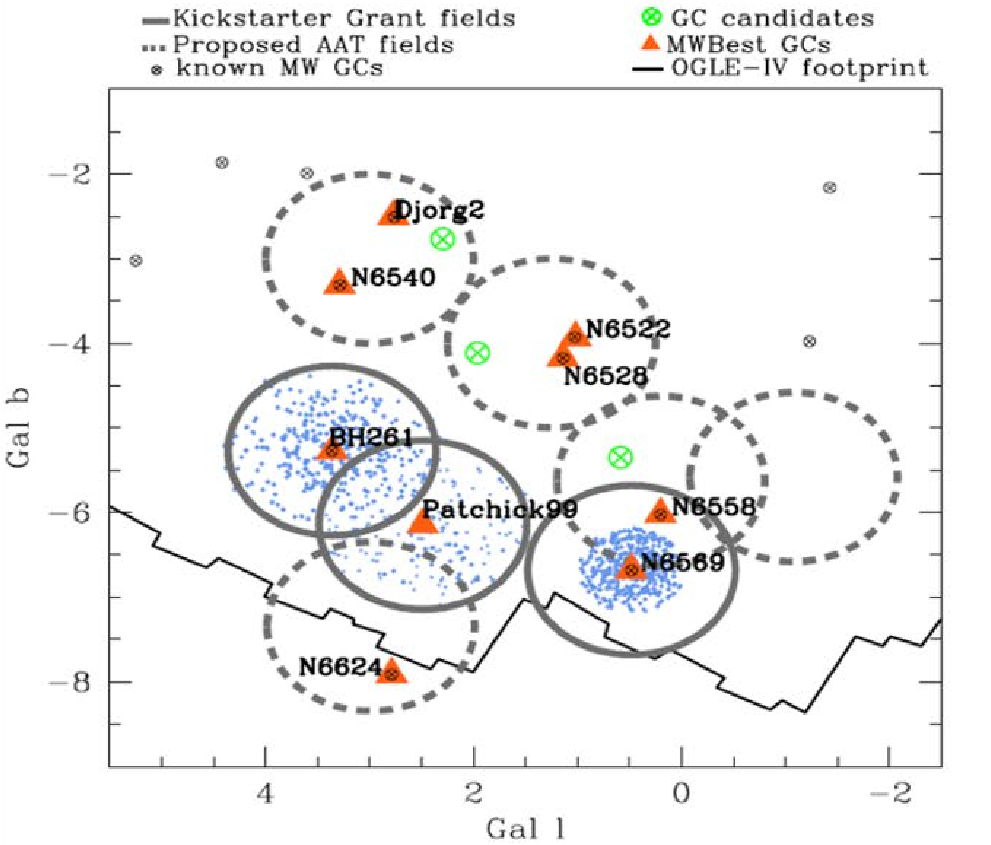
# Milky Way's Central Bulge Stars Formed in Single Burst of Formation over 10 Billion Years Ago – Christian Johnson (BDBS survey)



BDBS Survey: 250 million stars and > 20 globular clusters



These GCs have been confined to the bulge for billions of years: are they being tidally stripped until the stars are all part of the bulge field star population?




**Blanco DECam Bulge Survey (BDBS) IV: Metallicity Distributions and Bulge Structure from 2.6 Million Red Clump Stars**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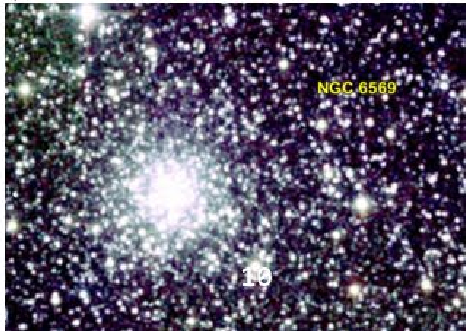
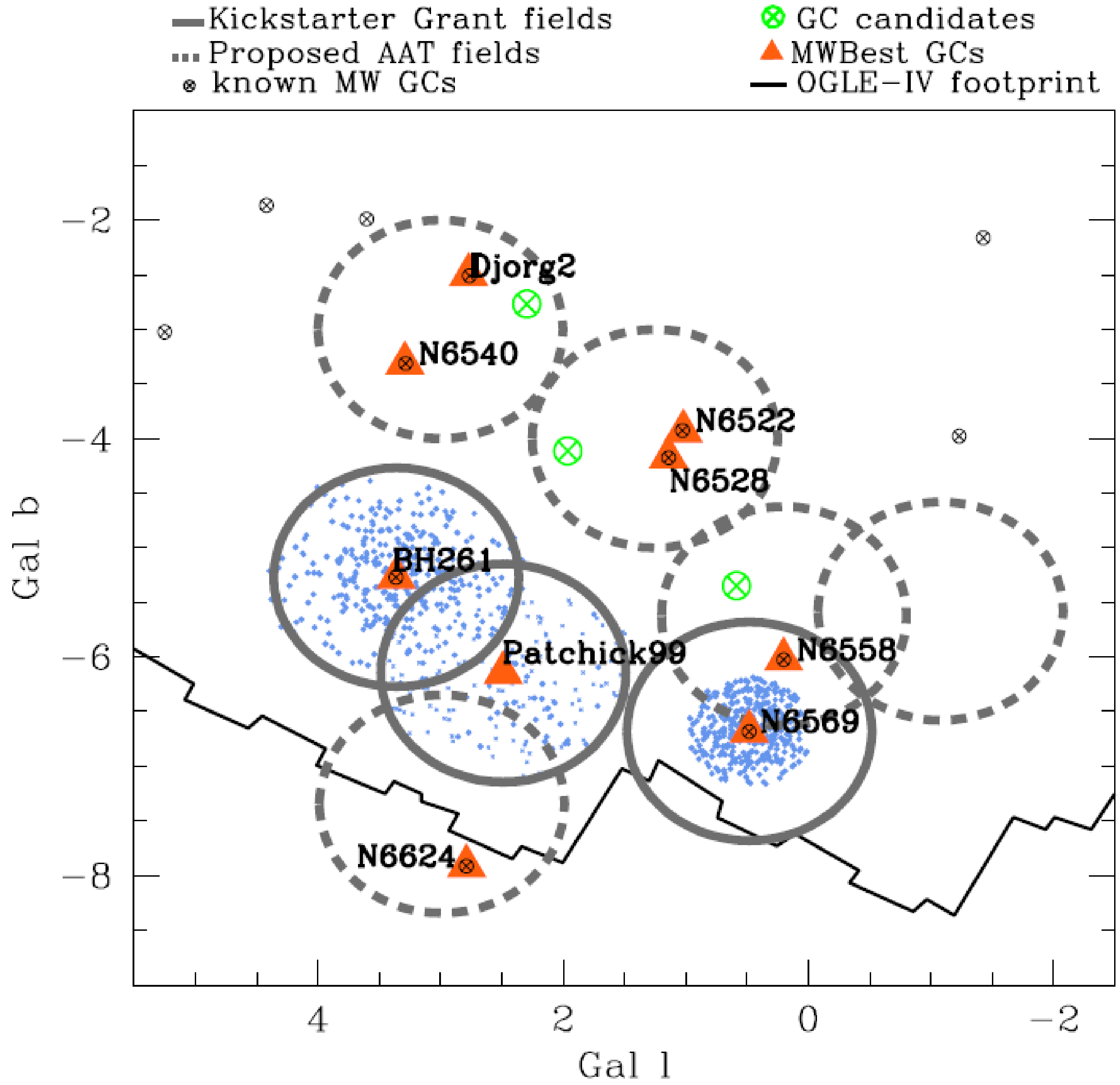
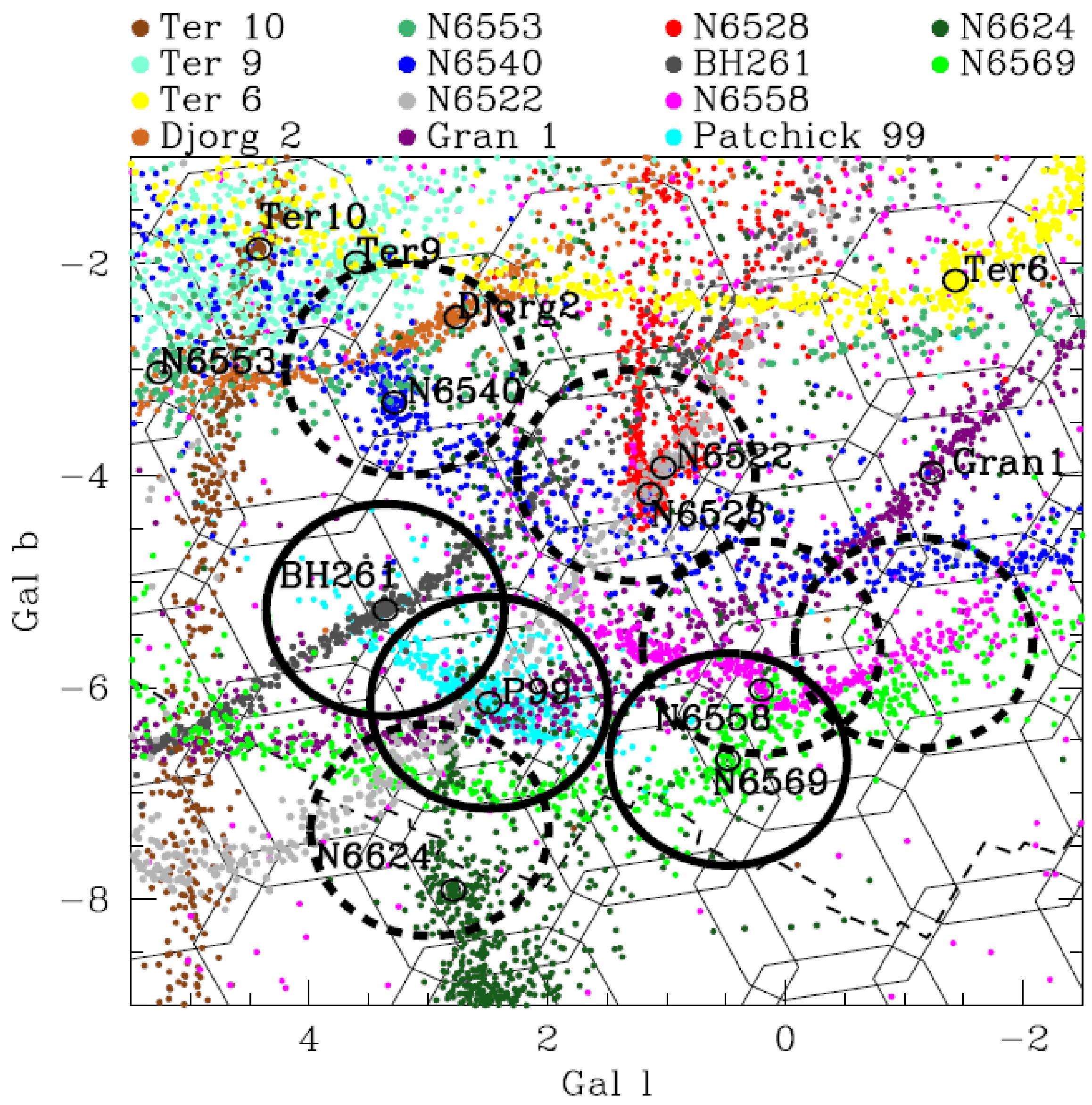
METHOD

- Pick ~20 poorly-studied GCs from the BDBS Survey
- Determine distance, size, age, metallicity, and membership-based on Gaia and BDBS data
- Search for extra-tidal stars, tidal tails, and streams in the databases and select stars for follow-up spectroscopy (AAT, APO, etc.)






## Gala-modeled likely tidal debris



# Measurements Needed

- Brightness of stars through several filters, spanning UV to IR wavelengths.
- Spectra of stars to obtain chemical composition and radial velocity.
- Positional information to measure stellar parallax (distance), and proper motion.

Stellar parallax is the apparent yearly shift of a nearby star against more distant background stars as Earth orbits the Sun. The parallax angle $p$ is half the total angular shift measured six months apart, and the distance is

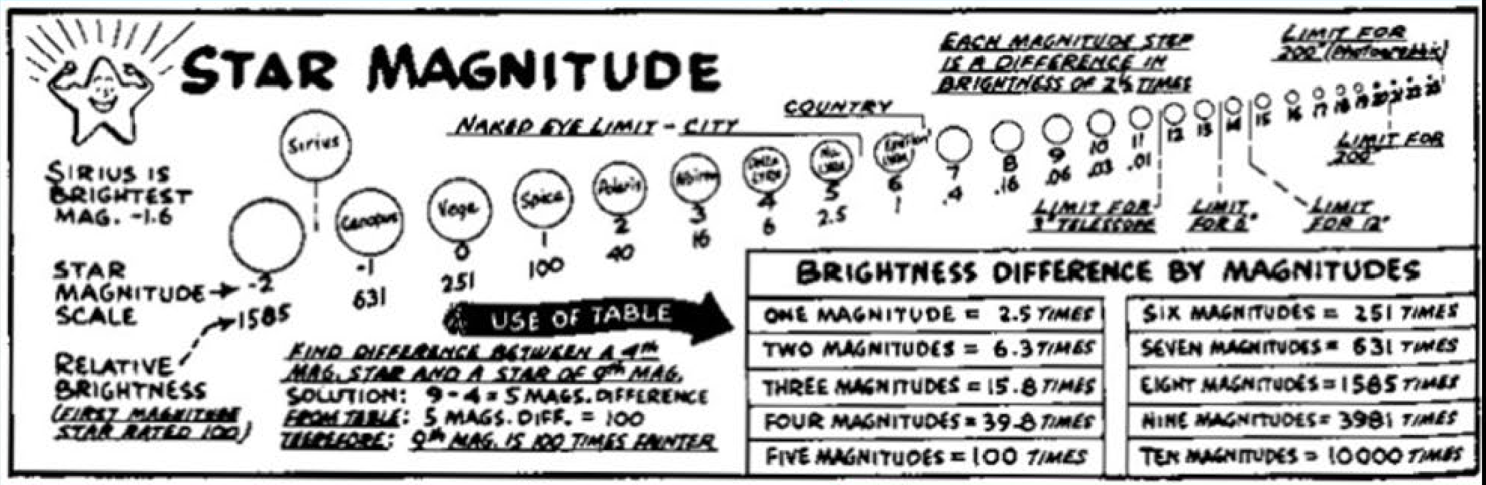
$$d(\mathrm{pc}) = \frac{1}{p(\mathrm{arcsec})}.$$

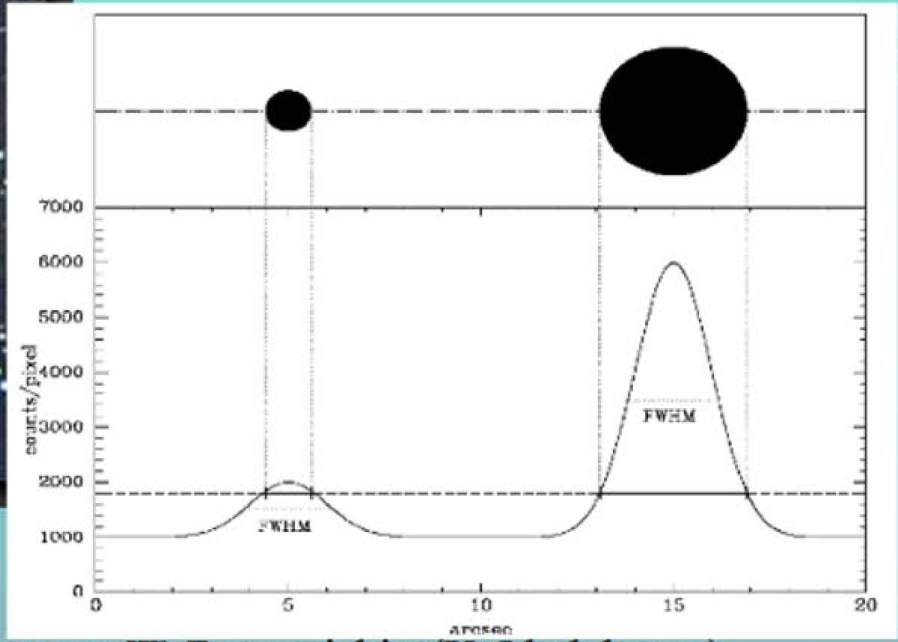
A star with a parallax of 1 arcsecond is therefore 1 parsec away.

Proper motion (PM) is the apparent angular motion of a star across the sky relative to distant background objects. It is usually measured in milliarcseconds per year and has two components: motion in right ascension and motion in declination.

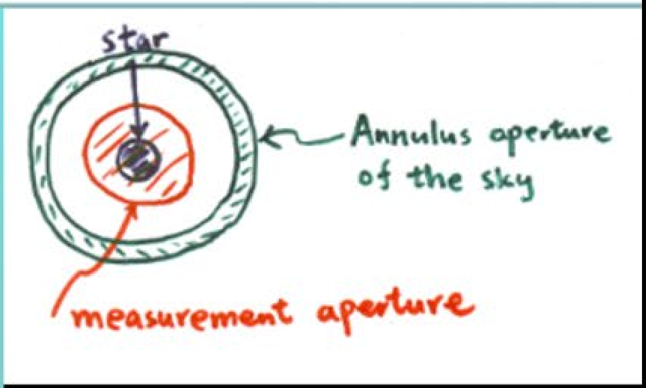
Radial velocity (RV) is the component of a star’s motion directly toward or away from the observer.
It is measured from the Doppler shift of spectral lines: a shift to longer wavelengths (redshift) indicates motion away from us, while a shift to shorter wavelengths (blueshift) indicates motion toward us.

# Photometry

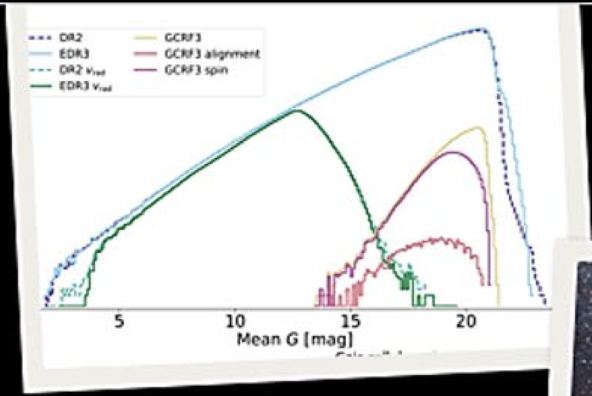




Courtesy: W. Romanishin (U Okalahoma)



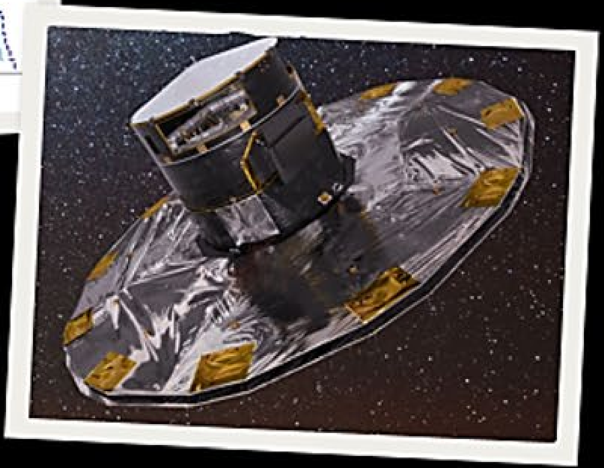

Imaging through various filters to find brightness and color (temperature) of stars

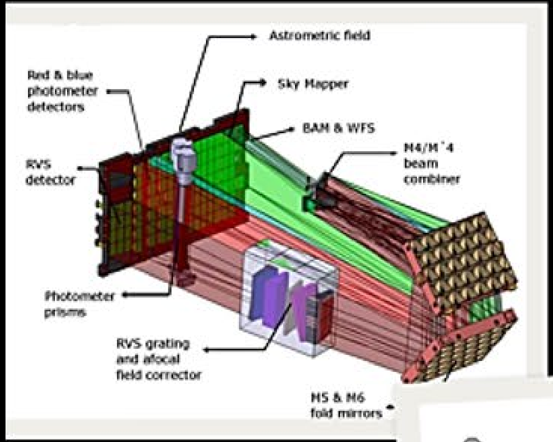
# GAIA: DISTANCES & MOTIONS

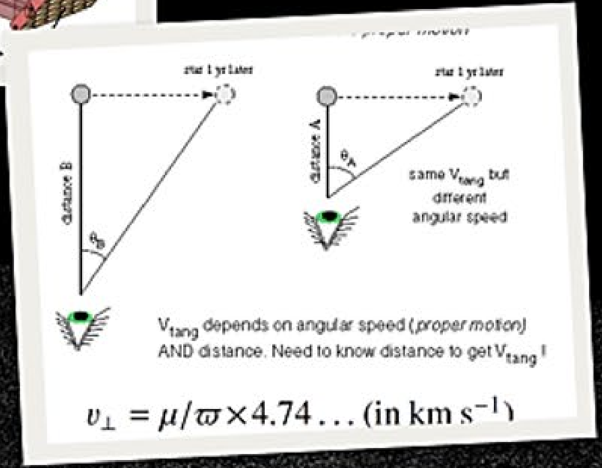



***Gaia*** is a space observatory of the European Space Agency (ESA), launched in 2013 and operated until early 2025. The spacecraft is designed for astrometry: measuring the positions, distances and motions of stars with unprecedented precision (Wikipedia and ESA).

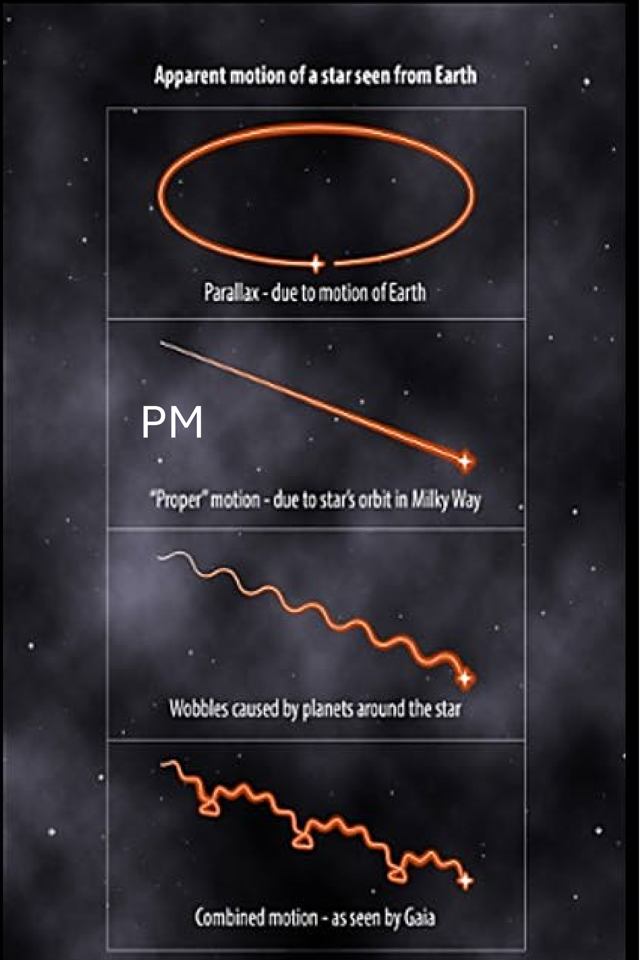
Gaia is a brightness-limited survey



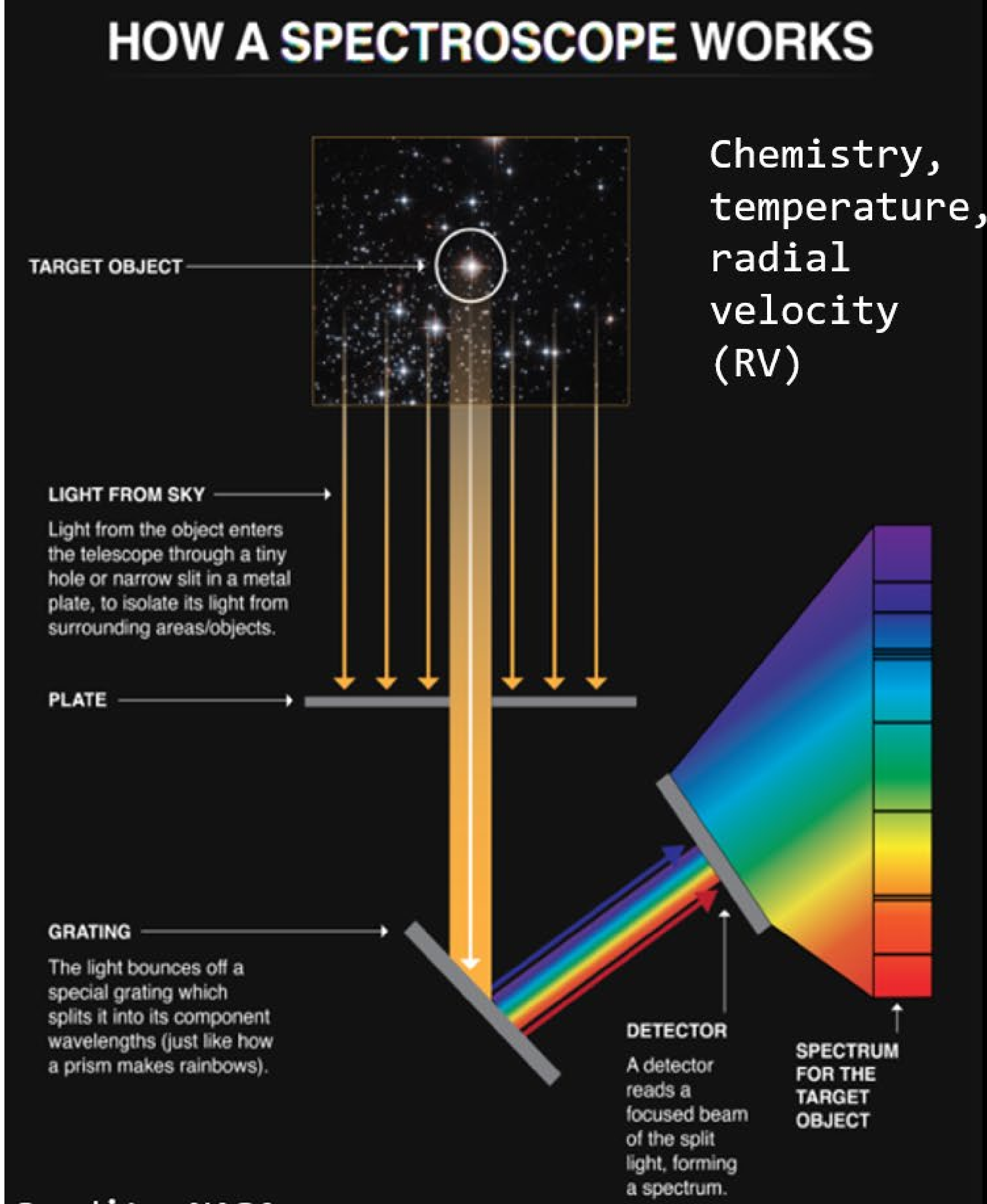



- Positions
- Parallaxes
- Proper Motions
- Photometry (G, Bp, and Rp)
- Temperatures (and other stellar params)
- Radial Velocities

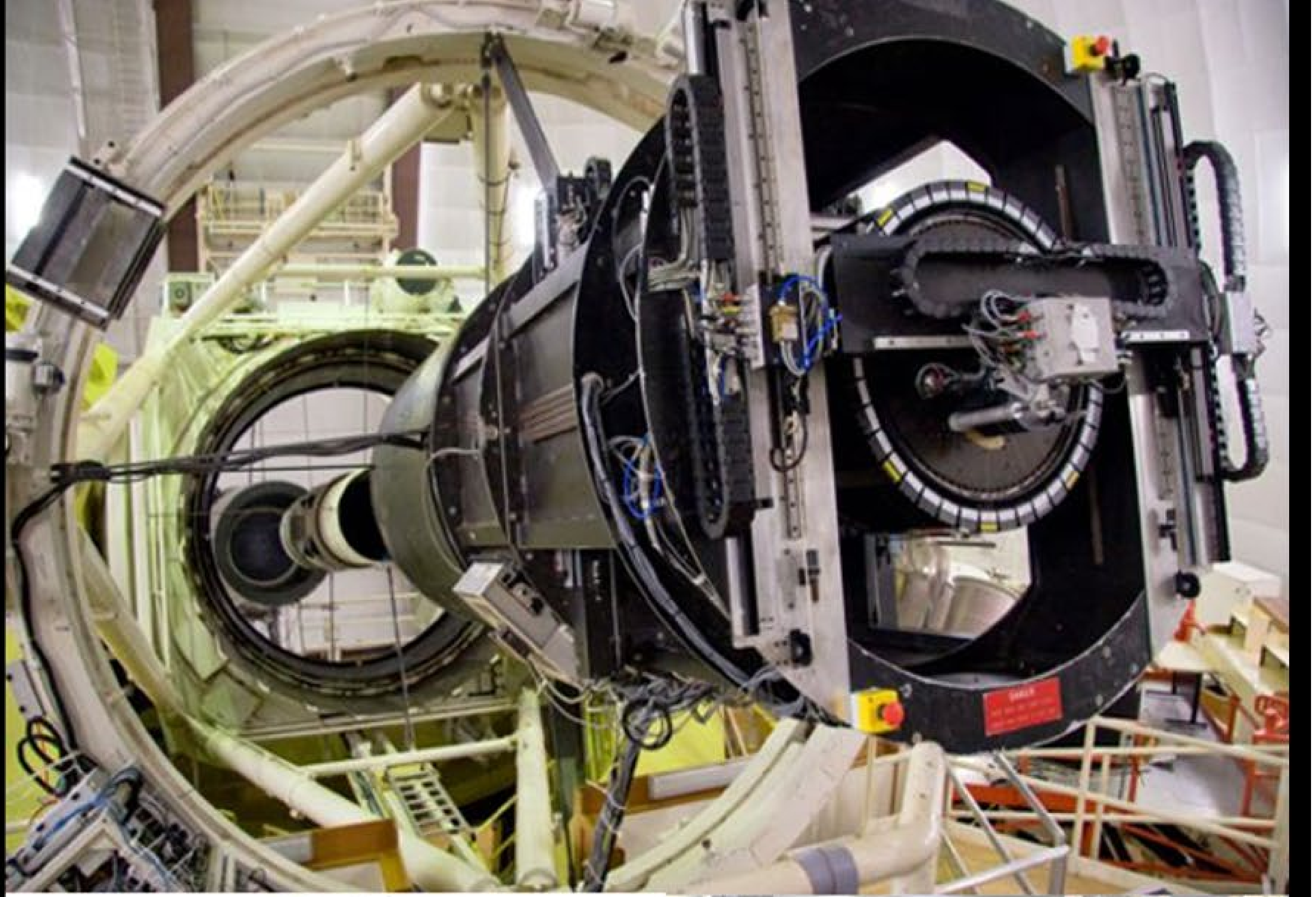


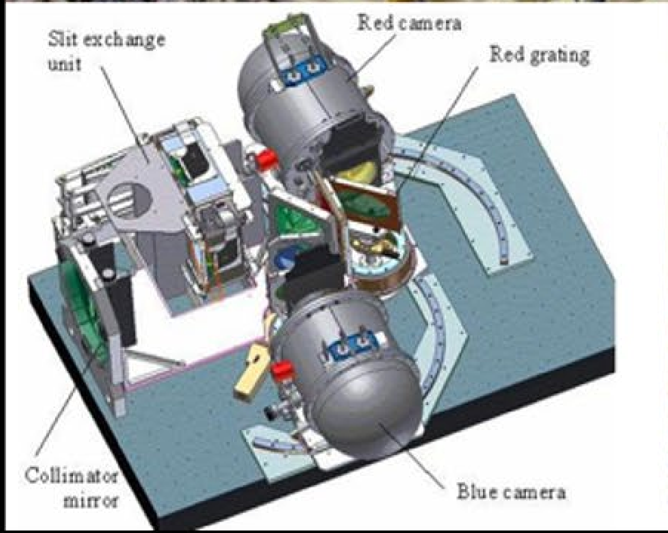

"When Gaia measures the distances to stars, it observes the combined effect of parallax motion and proper motion. Since parallax motion repeats every year, while proper motion is a continuous, linear effect, a few years of observational monitoring allow the Gaia scientists to separate the two effects for every star." - ESA

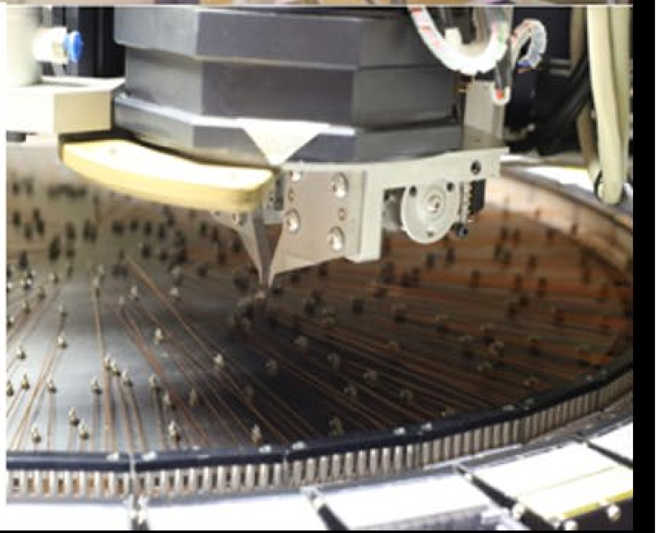


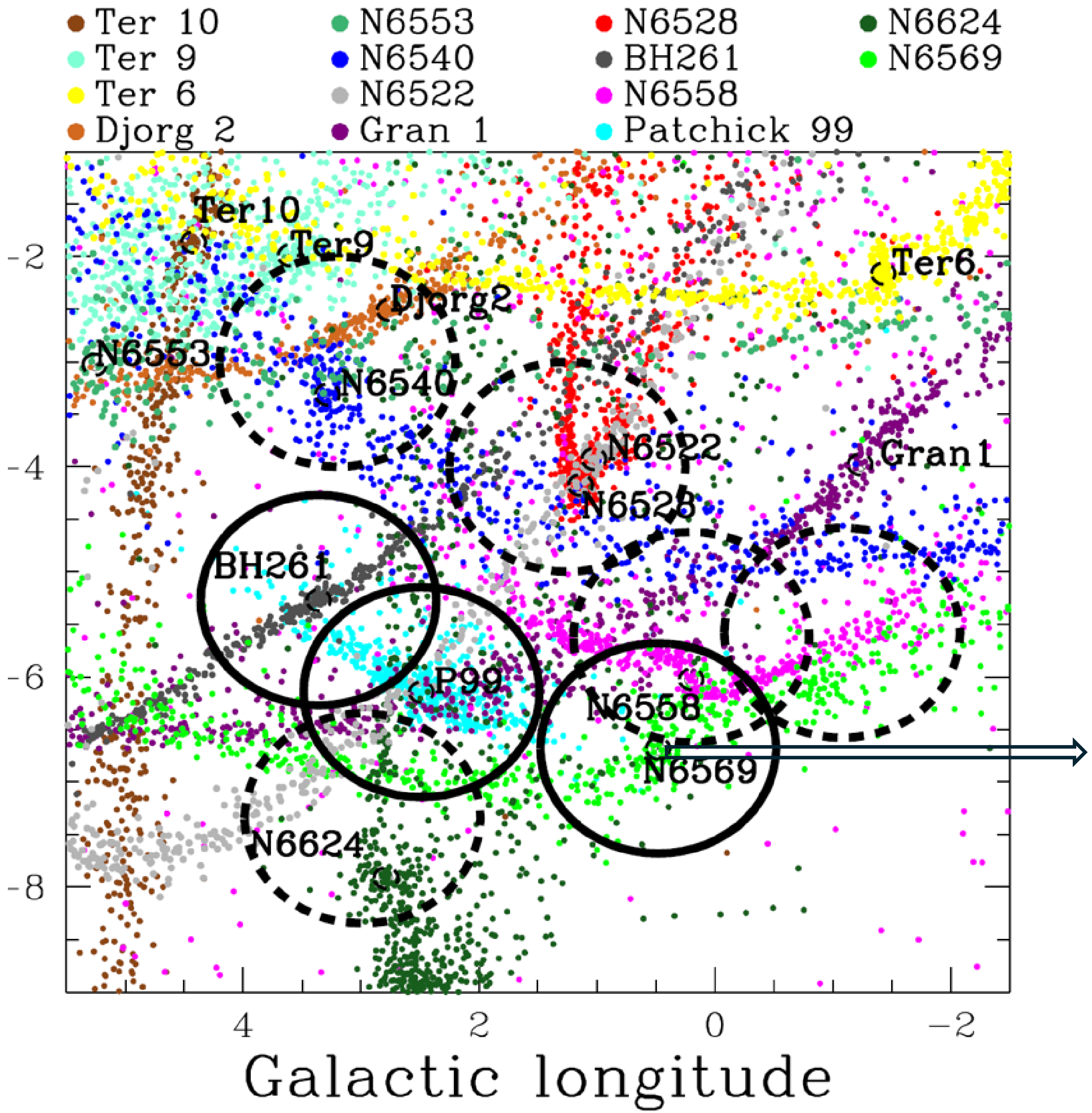

Chemistry, temperature, radial velocity (RV)

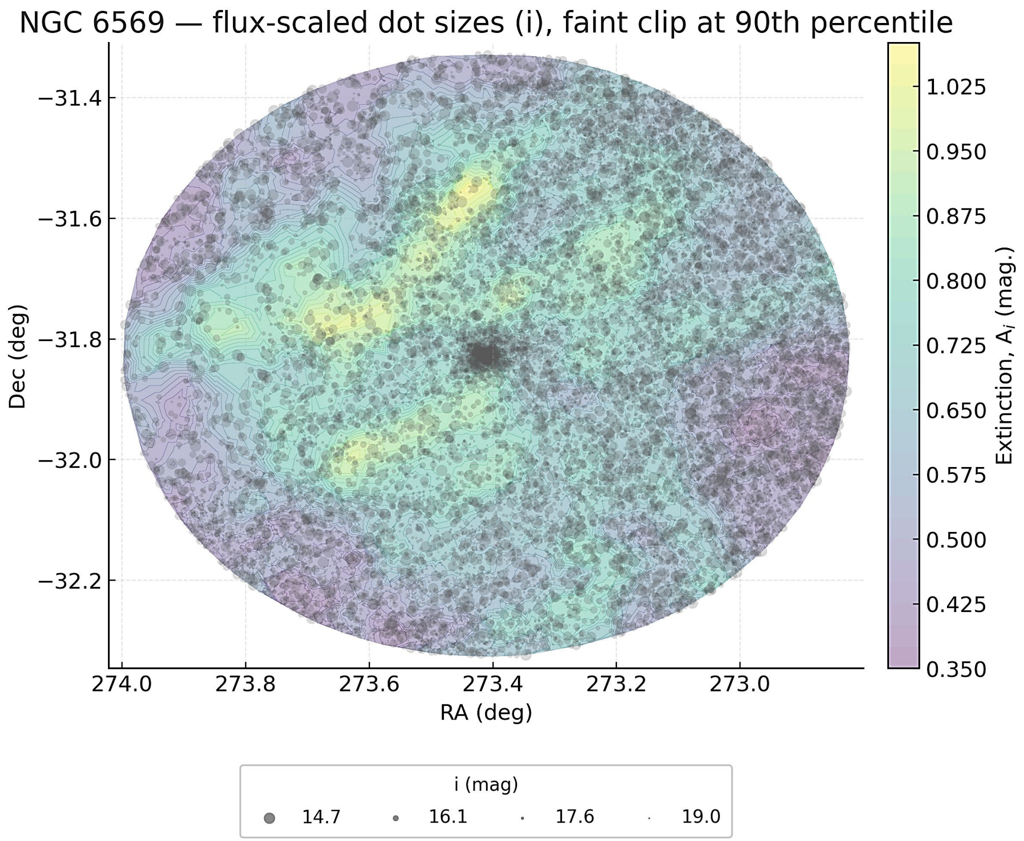
Credit: NASA

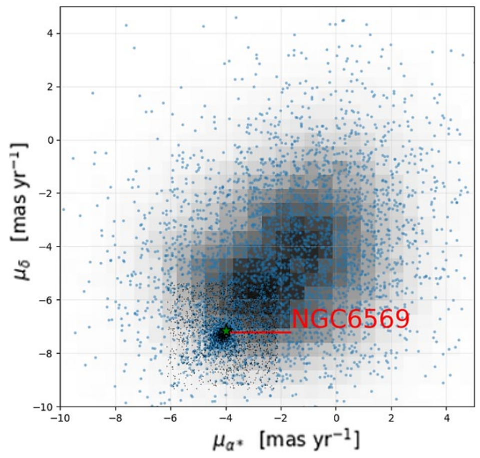





Ter 10
Ter 9
Ter 6
Djorg 2
N6553
N6540
N6522
Gran 1
N6528
BH261
N6558
Patchick 99
N6624
N6569
Ter10
Ter9
Djorg2
N6553
N6540
N6522
N6528
Ter6
Gran1
BH261
P99
N6558
N6569
N6624
-2
-4
-6
-8
4
2
0
−2
Galactic longitude


NGC 6569 — flux-scaled dot sizes (i), faint clip at 90th percentile
−31.4
−31.6
−31.8
−32.0
−32.2
Dec (deg)
274.0
273.8
273.6
273.4
273.2
273.0
RA (deg)
1.025
0.950
0.875
0.800
0.725
0.650
0.575
0.500
0.425
0.350
Extinction, $A_i$ (mag.)
i (mag)
14.7
16.1
17.6
19.0


NGC6569
$\mu_\delta$ [mas yr$^{-1}$]
$\mu_{\alpha*}$ [mas yr$^{-1}$]

# Red vs. Blue Spectral Regions

- Spectra from the blue part of the visual band contain more absorption lines of the elements which are able to give the chemical composition, surface temperature, and physical state of the stars, as well as radial velocities.
- Spectra from the red part of the spectrum contain the calcium II triplet, which is extremely strong in relation to other absorption lines, and can be detected in very faint stars. It can be used for "metallicity" and radial velocities.
- Red spectra are also more useful in regions affected by interstellar dust extinction.

# Stellar Iron Abundance: [Fe/H]

The iron abundance of a star relative to the Sun is written as

$$[\mathrm{Fe}/\mathrm{H}] = \log_{10}\left(\frac{N_{\mathrm{Fe}}}{N_{\mathrm{H}}}\right)_{\mathrm{star}} - \log_{10}\left(\frac{N_{\mathrm{Fe}}}{N_{\mathrm{H}}}\right)_{\odot},$$

where $N_{\mathrm{Fe}}$ and $N_{\mathrm{H}}$ are the numbers of iron and hydrogen atoms.

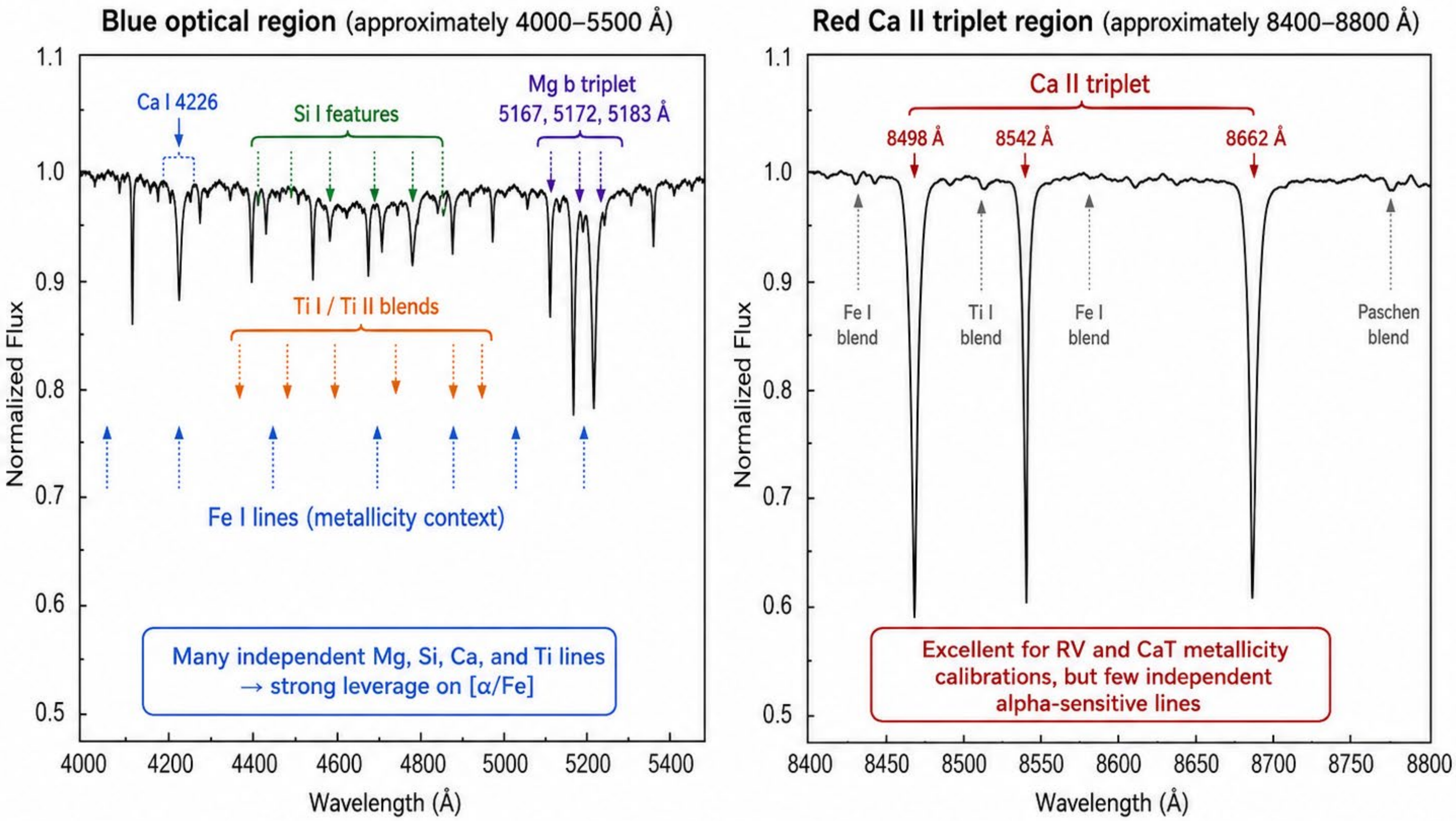
A star with $[\mathrm{Fe}/\mathrm{H}] = 0$ has the same iron-to-hydrogen ratio as the Sun. A star with $[\mathrm{Fe}/\mathrm{H}] = -1$ has one-tenth the solar iron abundance, while a star with $[\mathrm{Fe}/\mathrm{H}] = -2$ has one-hundredth the solar iron abundance.

The square-bracket abundance notation is generally credited to George Wallerstein and appeared in a 1959 paper by H. L. Helfer, George Wallerstein, and Jesse Greenstein. It is therefore safest to say that Wallerstein helped introduce and establish the notation, rather than that he was necessarily its sole inventor.

**Metals** are all elements heavier than helium. In astronomy, a star's metallicity usually refers to the abundance of heavy elements relative to hydrogen, commonly measured using $[\mathrm{Fe}/\mathrm{H}]$.

**Alpha elements** are metals mainly produced by adding helium nuclei, or alpha particles, during stellar nucleosynthesis. Important alpha elements include oxygen, neon, magnesium, silicon, sulfur, argon, calcium, and titanium. Their abundance relative to iron is written as $[\alpha/\mathrm{Fe}]$.

# Why Blue Spectra Constrain [α/Fe] Better Than Red CaT Spectra




**Blue spectra contain many independent alpha-element diagnostics.
Red CaT spectra are dominated by the Ca II triplet,
so [α/Fe] is much less well constrained, especially at moderate resolution and S/N.**

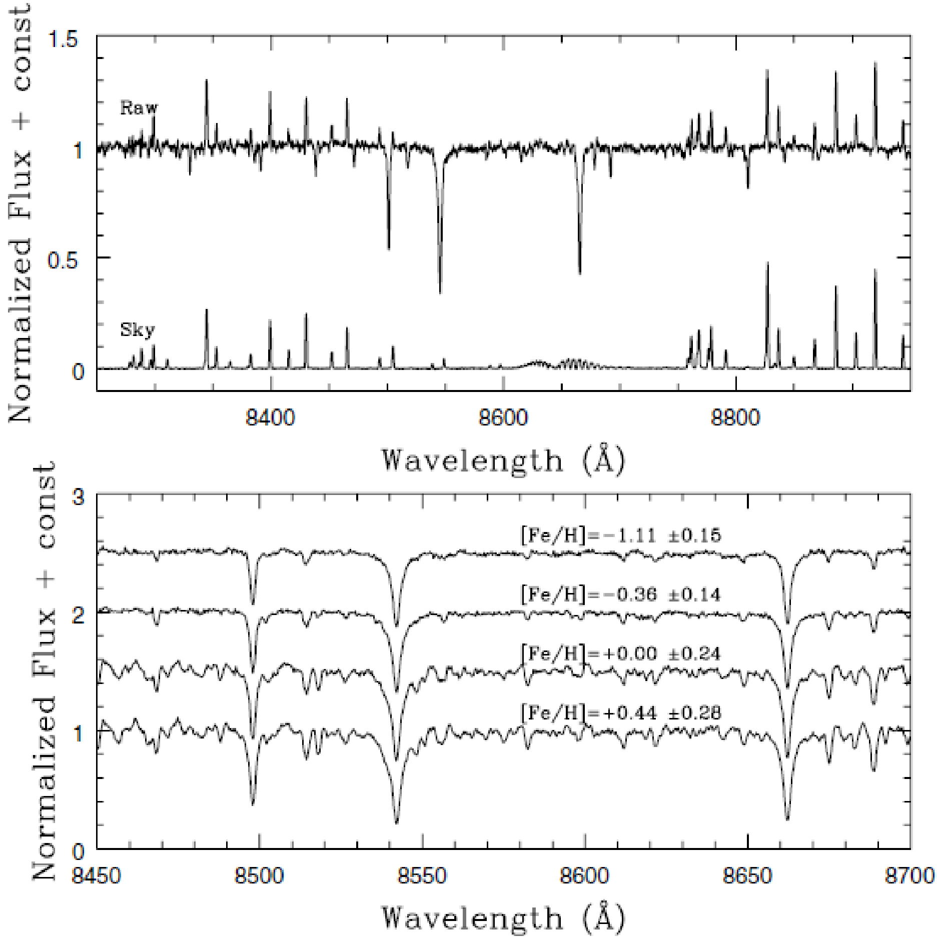



**Fig. 2.** *Upper panel*: normalised raw spectrum for a typical Milky Way bulge star together with the corresponding sky spectrum. The *lower panel* shows processed spectra for four bulge stars spanning the typical metallicity range of the Galactic bulge. The increasing strength of the CaT lines with metallicity is evident, as is the increasing number of small atomic lines contaminating the continuum.

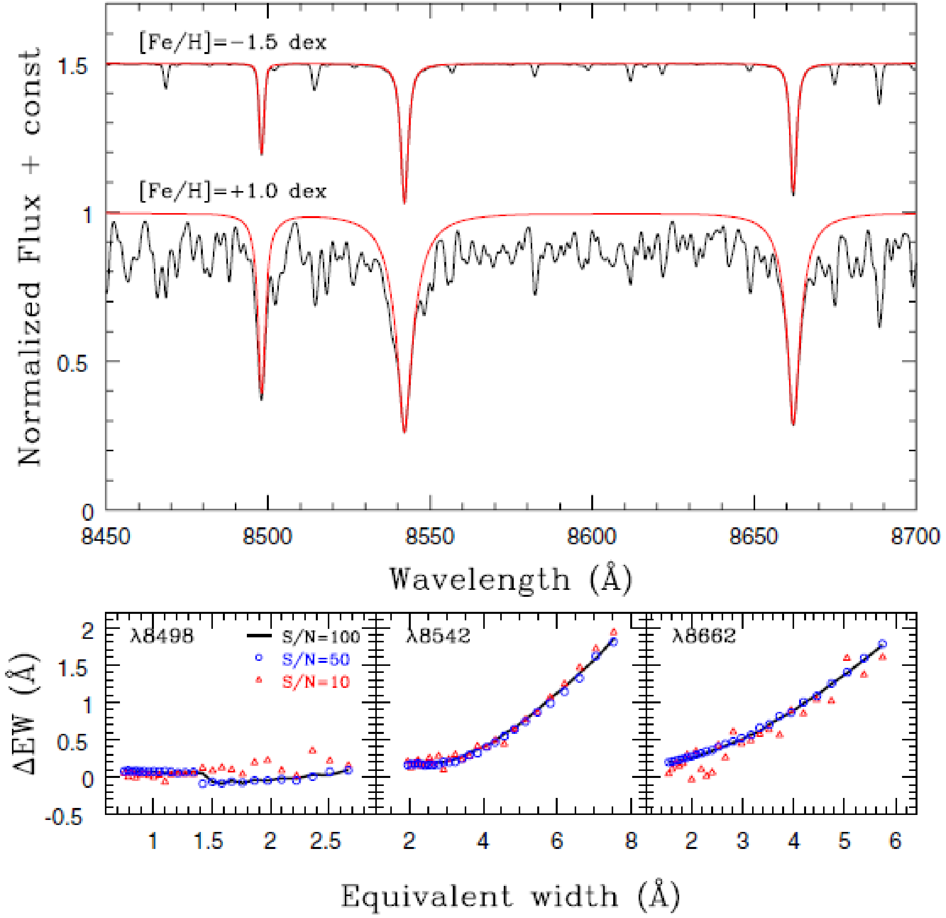
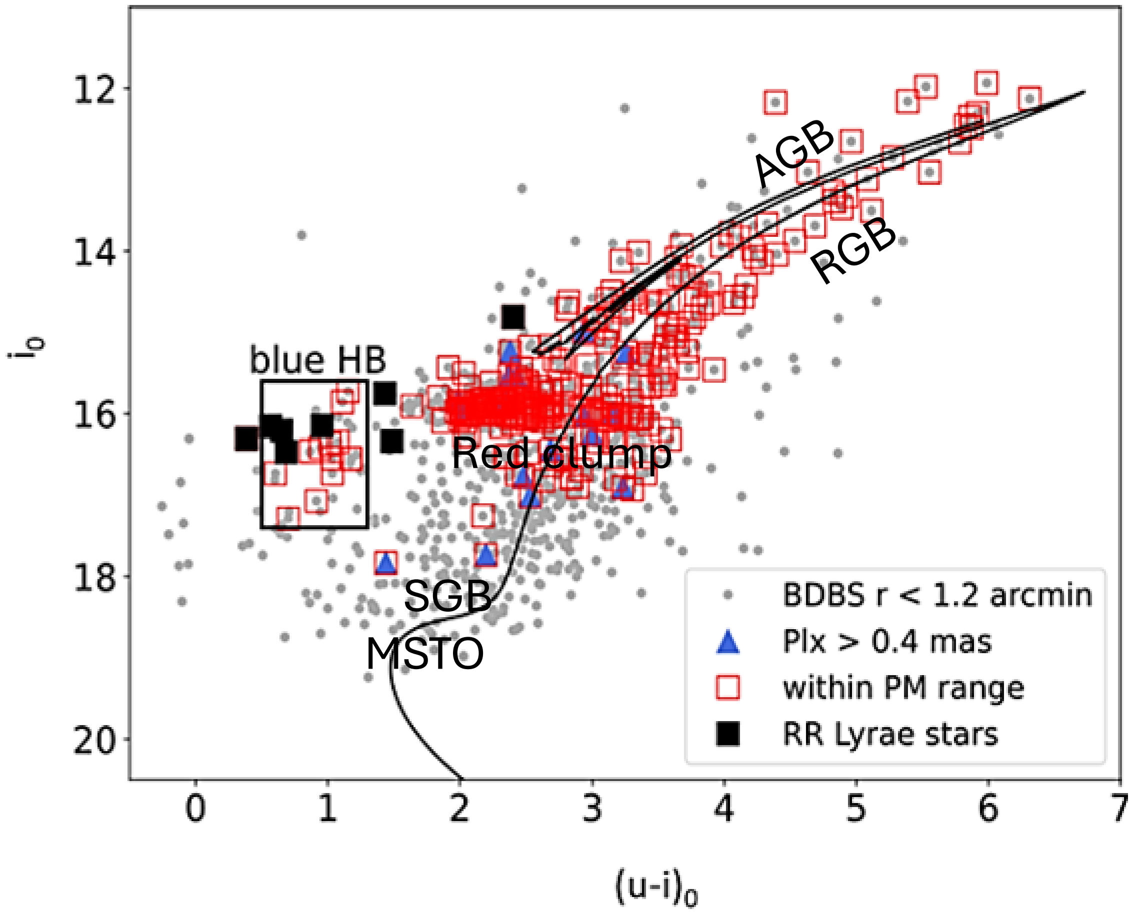
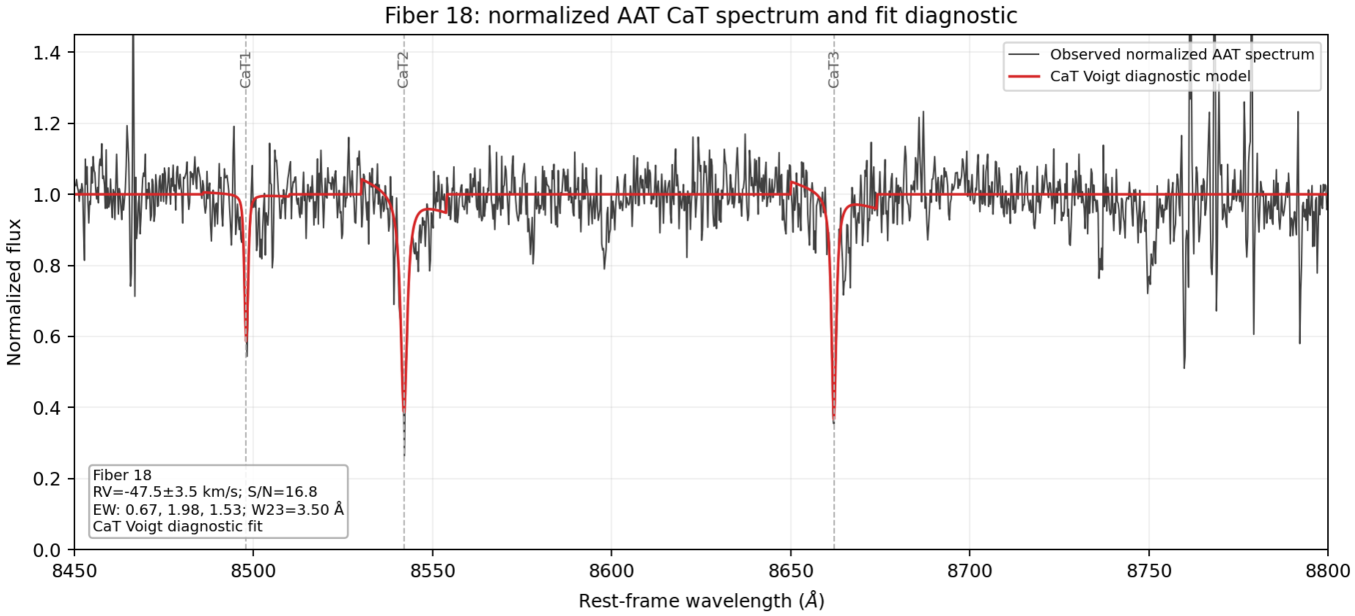


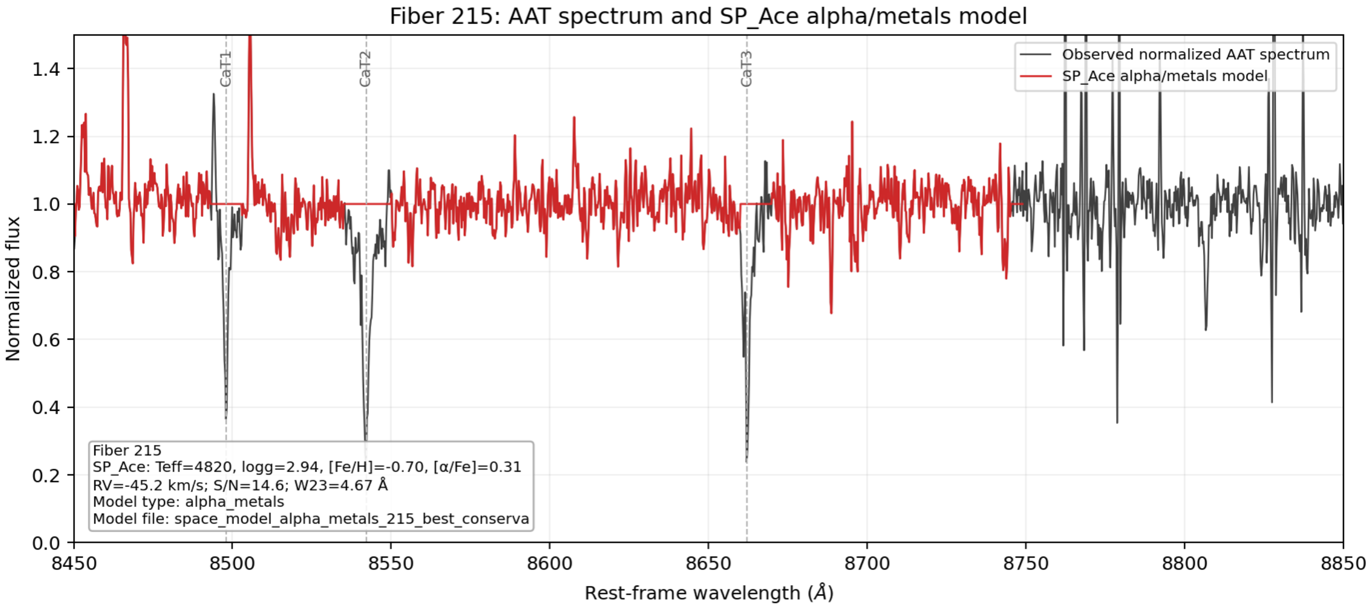

**Fig. 3.** *Upper panel*: comparison between the more metal poor and metal rich synthetic spectra from our library, in the region around the CaT lines. In both cases, the black line shows the spectra including the lines of all elements, while the red lines shows only the three CaT lines. The *lower panels* show the difference between the *real* EW, as measured in the red spectra, and the *measured* EW, as measured in the black spectra, for each of the three CaT lines as a function of EW and for three values of S/N.



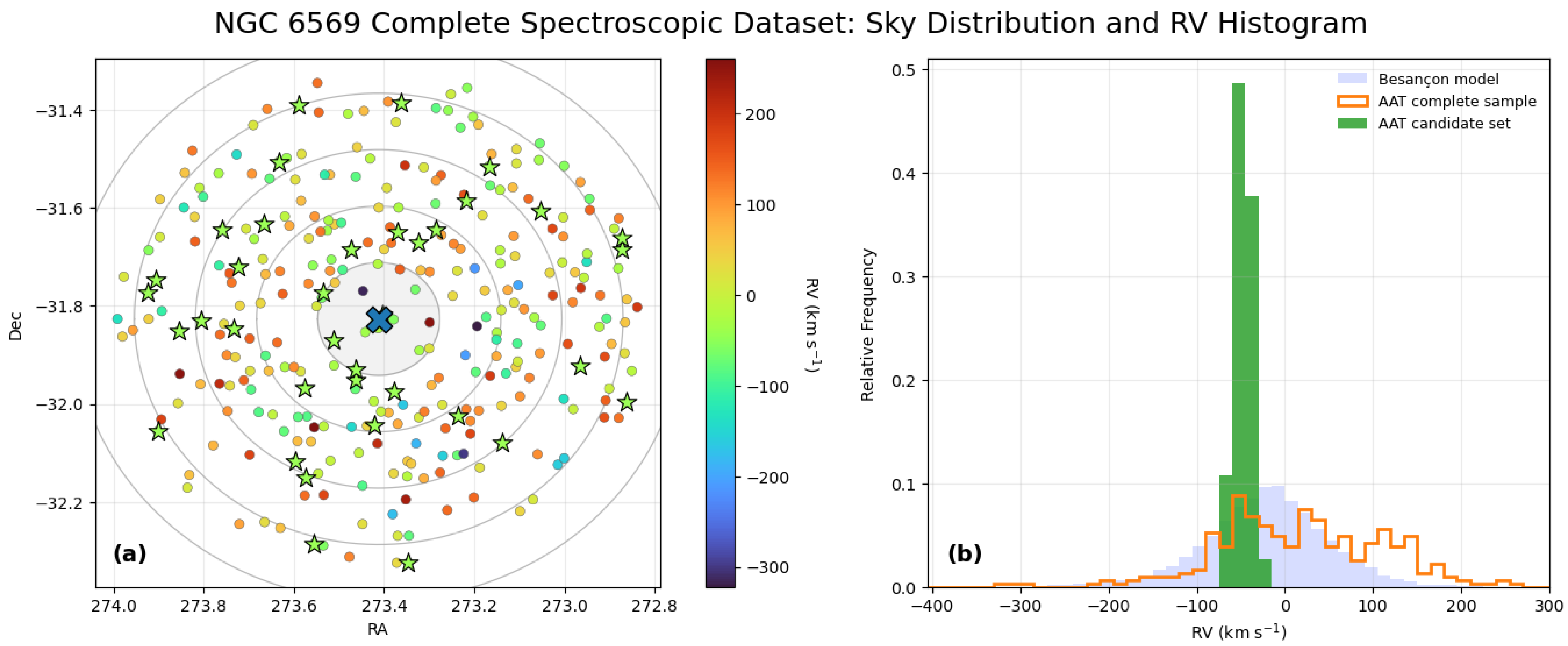

The BDBS dereddened CMD for is shown for i vs. (u-i), dereddened using the BDBS reddening maps (Simion et al. 2017). The small, gray, filled circles are the stars within 1.2' of the center of NGC 6569. The following objects are identified within 5 tidal radii of the center of the cluster: stars must have a proper motion within 1 mas/yr in R.A. and within 1.5 mas/yr in Dec of the main cluster proper motion: (~ 4.125 mas/yr, ~ 7.354 mas/yr). Additional constraints were applied: objects with BDBS u-magnitudes with uncertainties < 0.04 mag. were selected if they had detections in the BDBS g- and i-bands also. The red open squares were identified as targets for AAT/AAOmega spectroscopy. The blue filled triangles are flagged as likely foreground objects, and the black filled squares were identified as RR Lyrae stars. The isochrone shown is from the MIST models (Choi et al. 2016), scaled to be α-enhanced: [Fe/H] = −0.75 dex, [α/F e] = +0.3 dex, with an age of 13 Gyr.




Normalized continuum, with Voight fits to the CaT lines.



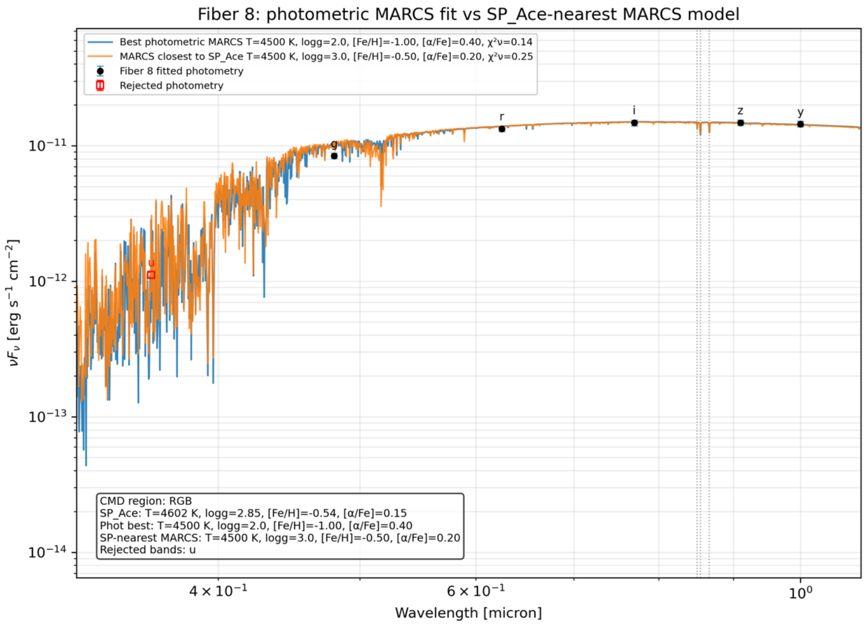

Normalized continuum, with SP_Ace fits to the regions outside the CAT lines, and avoiding residual sky emission lines.

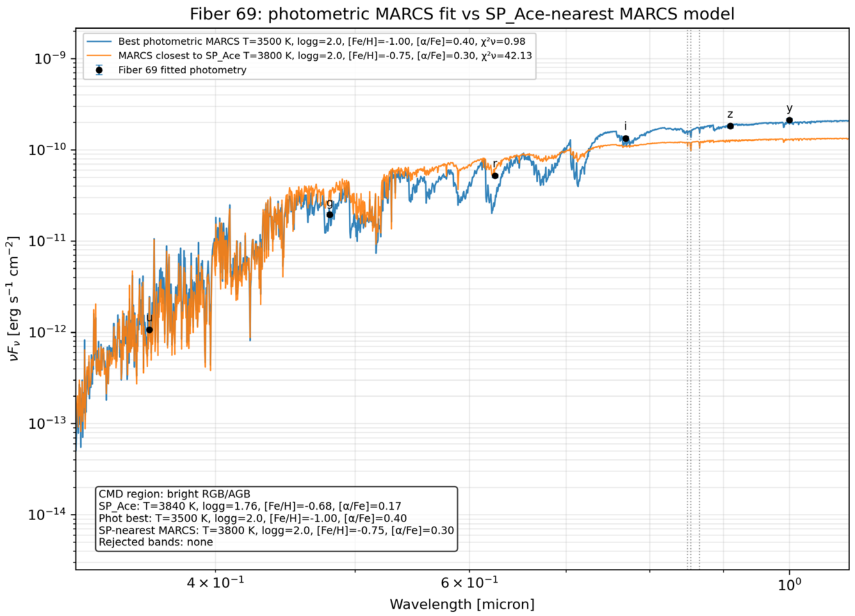


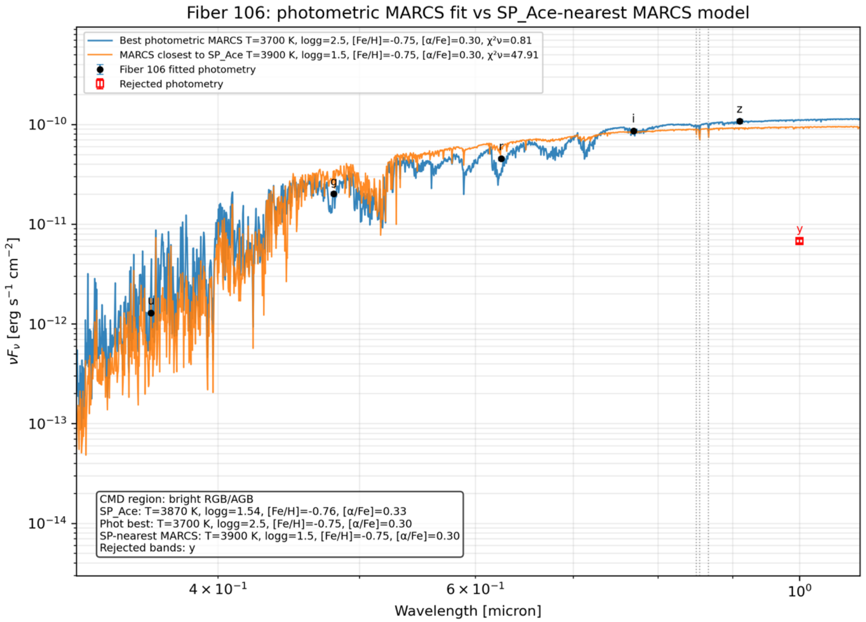

☆ RV-Candidate Escapees

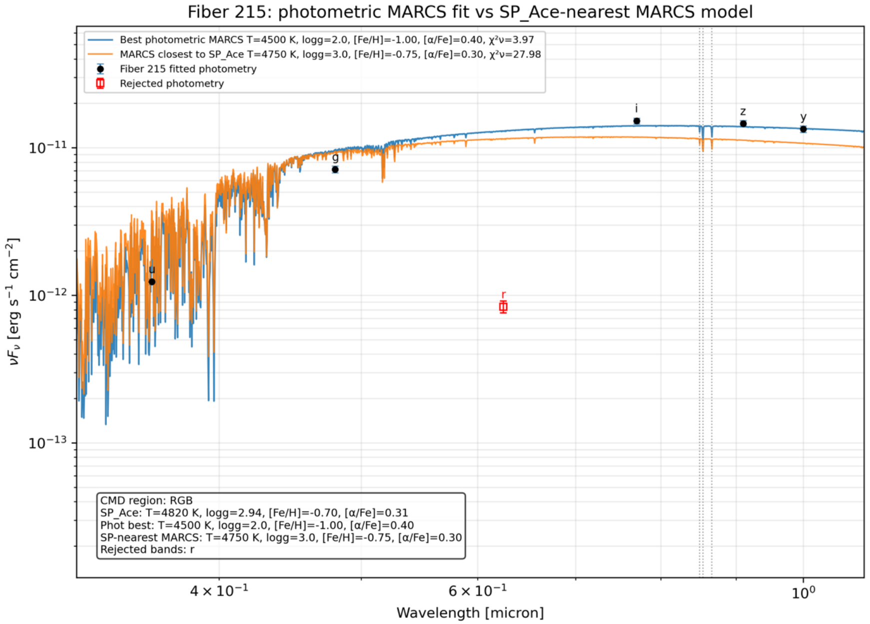
AAT complete sample: 304
Current paper-style candidates: 37
Besançon comparison stars: 7562

(a) A map of the R.A. versus decl. positions of the 304 spectroscopic targets We choose our search limit as −25 > RV > −65 km s-1, yielding 57 objects (black-edged green stars. (b) The relative number of stars per bin is shown in this histogram. The cluster's RV range yielded by Johnson et al. (2020), from −30 > RV > −63 km/s.



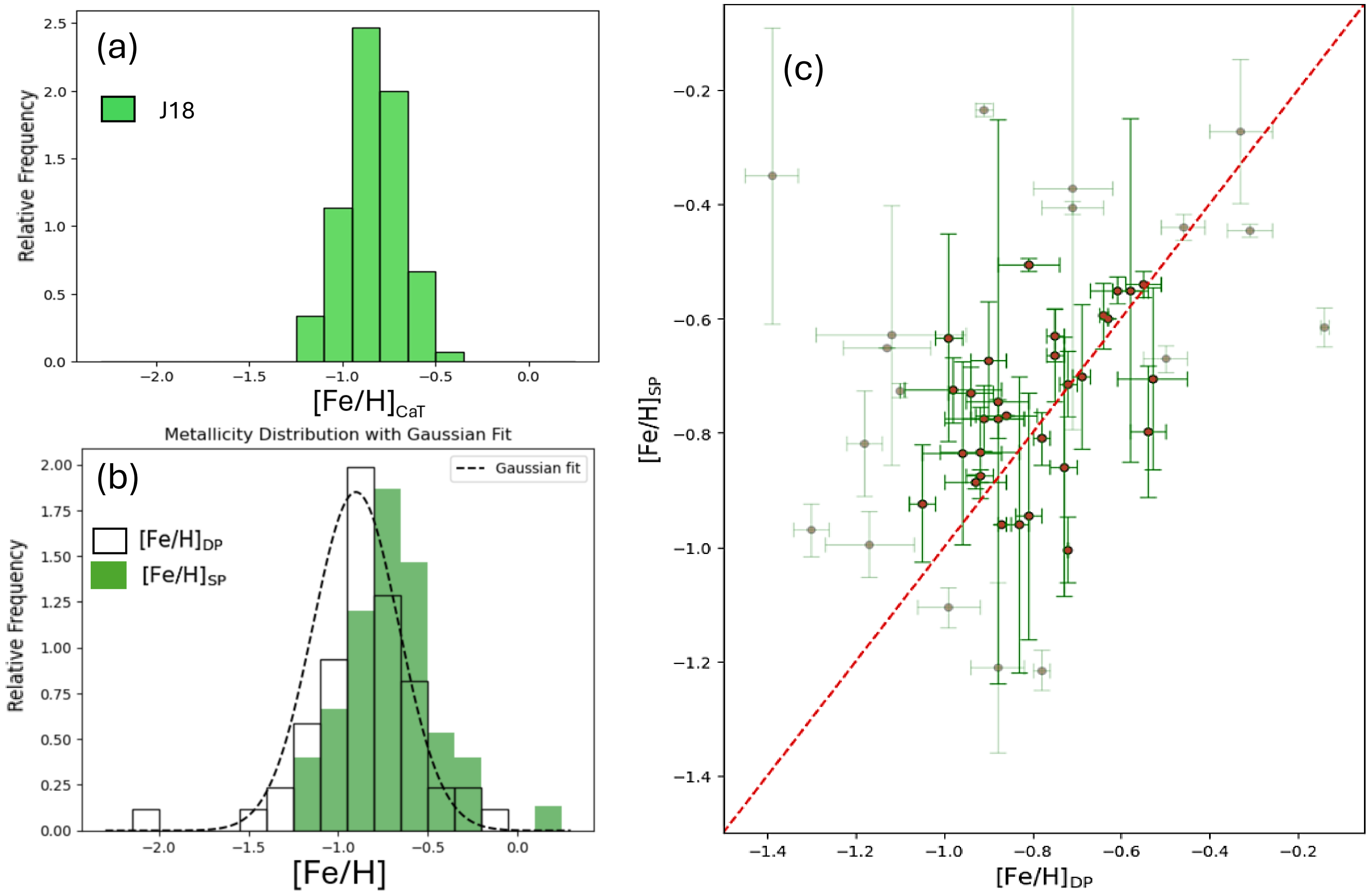




**Sanity Check:**

Photometry data fit to SED and spectral fits.




The source-density histograms:
(a) A histogram of the J18 sample of 98/100 stars is shown in green, which were matched with BDBS photometry. For these 98 stars, using the luminosity correction in the i-band, we find:

```
[Fe/H]J18 = -0.84 ± 0.17 dex
```

(b) The 58 stars in the RV-selected sample (-25>RV>-65 $kms^{-1}$ ) are shown as the black-line histogram (no fill), calibrated using Dias & Parisi's (2020) method.

```
[Fe/H]DP = -0.91 ± 0.4 dex
```

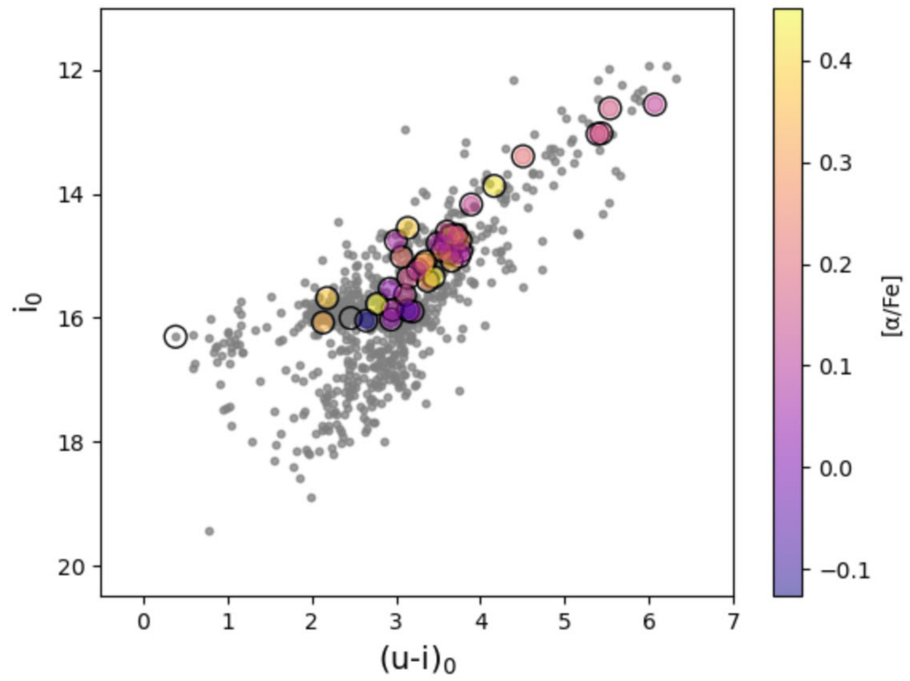
The objects with 3500<T(K) <7500 were fit using the SP_Ace code:

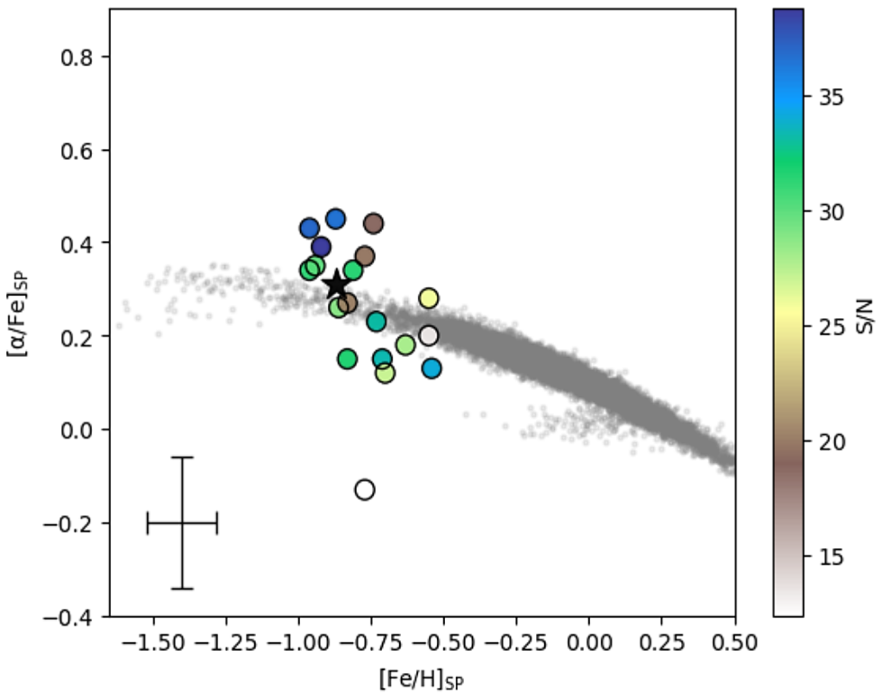
```
[Fe/H]SP = -0.78 ± 0.2 dex
```

(c) We compare the two fits for the overlap between the samples (all stars are faint green and orange filled circles). The red line is 1:1, which fits the stars with both $[Fe/H]_{DP}$ and $[Fe/H]_{SP}$ are limited to between -0.5 and -1.1 dex (brighter green and orange points).

The dotted line is the Gaussian for to the P23 data, presented in Crociati et al. (2023):

[Fe/H]P23 = -0.90±0.24 dex

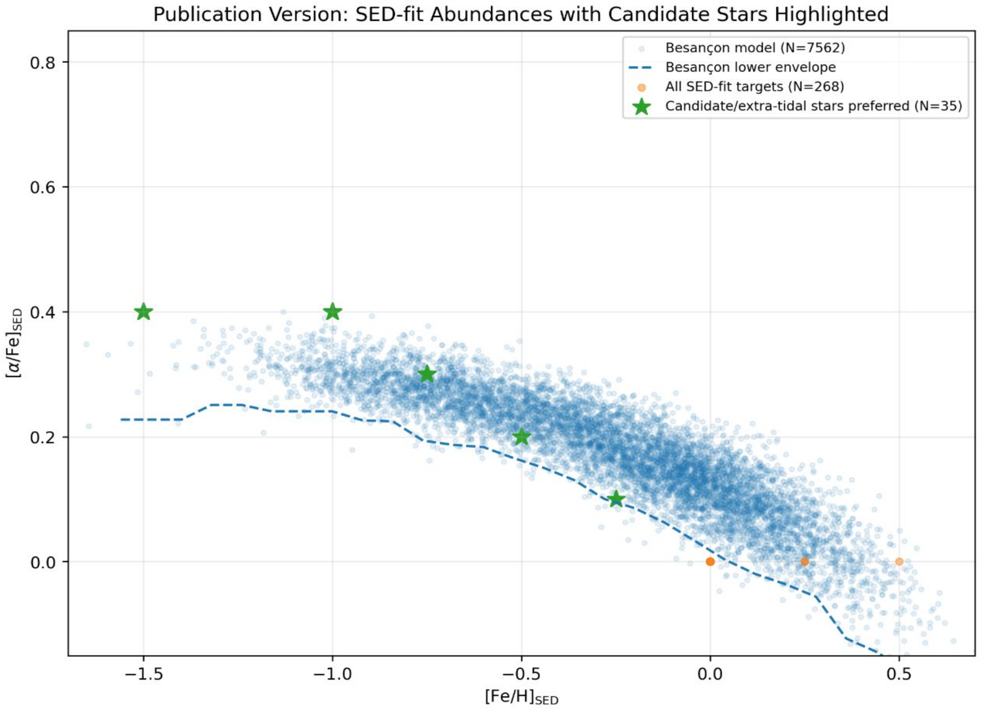


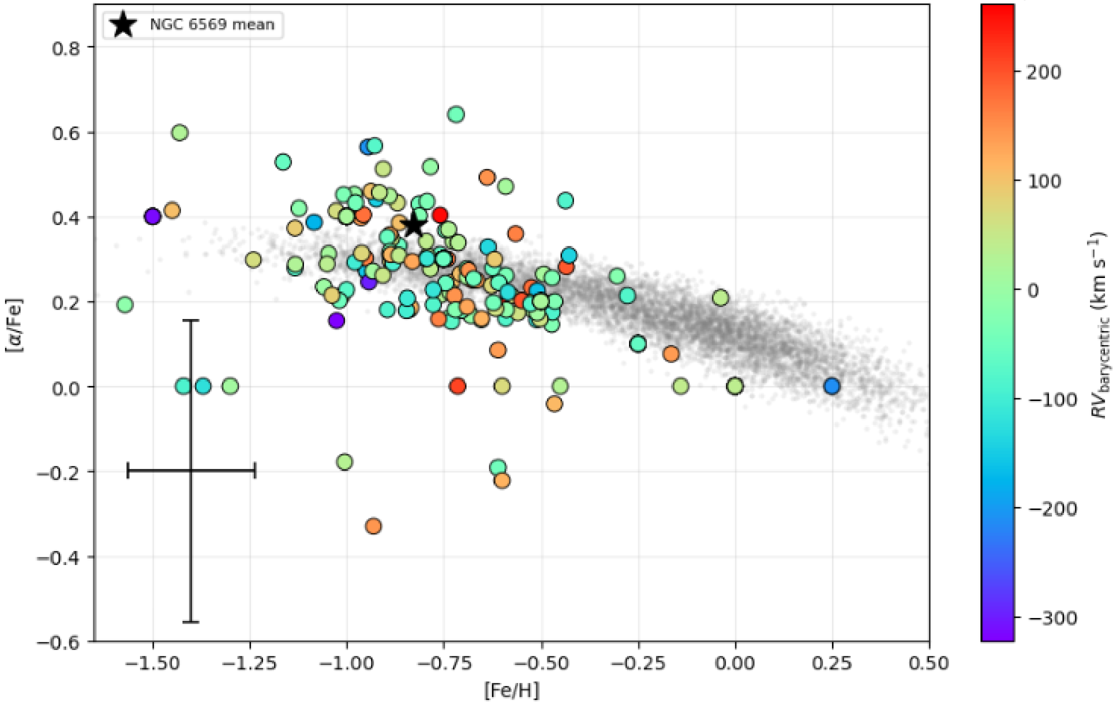




Top panel: 50 stars were within the RV range, which we show as black open circles, graded A-D in Table 4 of **Hughes et al. (2026).** Where we could measure $[\alpha/Fe]_{SP,}$ we shade the circles in with the color-coding on the right-hand bar.

The Tinsley-Wallerstein diagram against the Bescancon models in our FOV and selection criteria, for the A+B graded stars are shown as filled circles, color-coded for S/N. We show the average J18 values for the cluster as the filled black star. The average uncertainties are shown as the error bars on the point in the lower right side. Any object below S/N of 20 has a lower limit on the $[\alpha/Fe]_{SP}$ value unless it has:

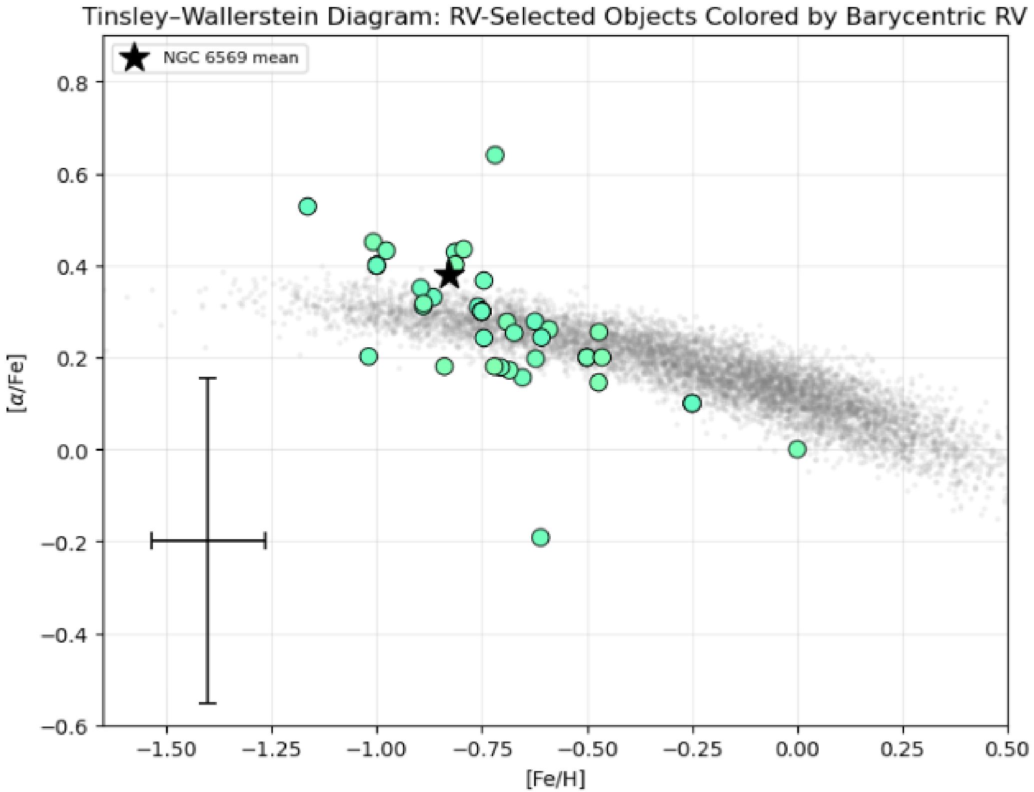
$$[\alpha/Fe]_{SP} > +0.15 \text{ dex}$$

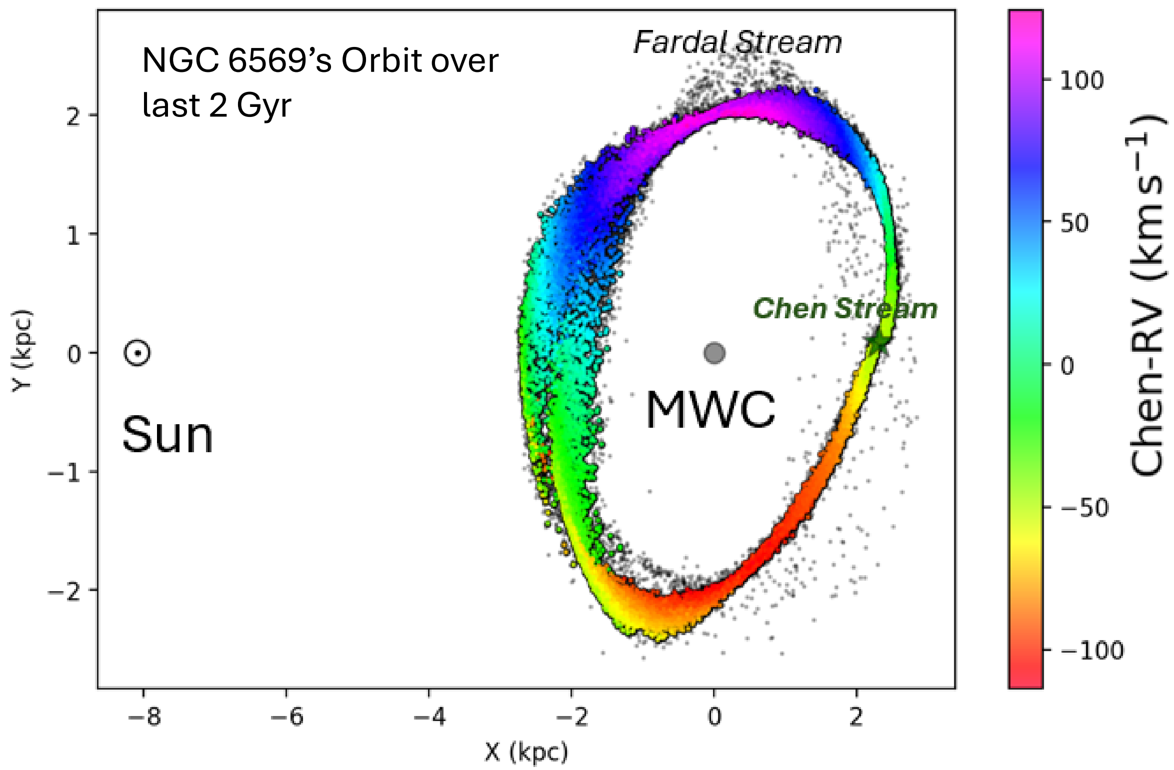


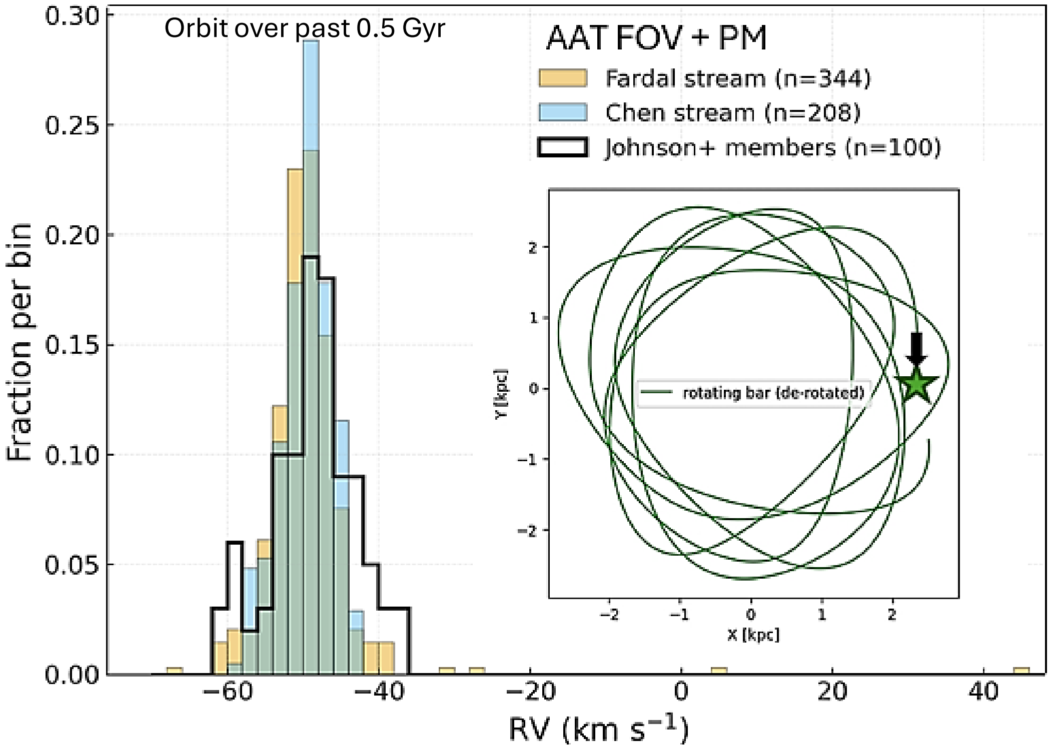

Re-analysis with the Voight line profiles and SED fits has not changed the results significantly.

```
All fibers with finite abundance + RV: 282
RV-selected fibers (-65 to -25 km/s): 51
Shared RV color scale:
  rv_min = -322.611
  rv_max = 260.56
Average e_FeH_adopted: 0.165
Average e_aFe_best_rule: 0.356
```

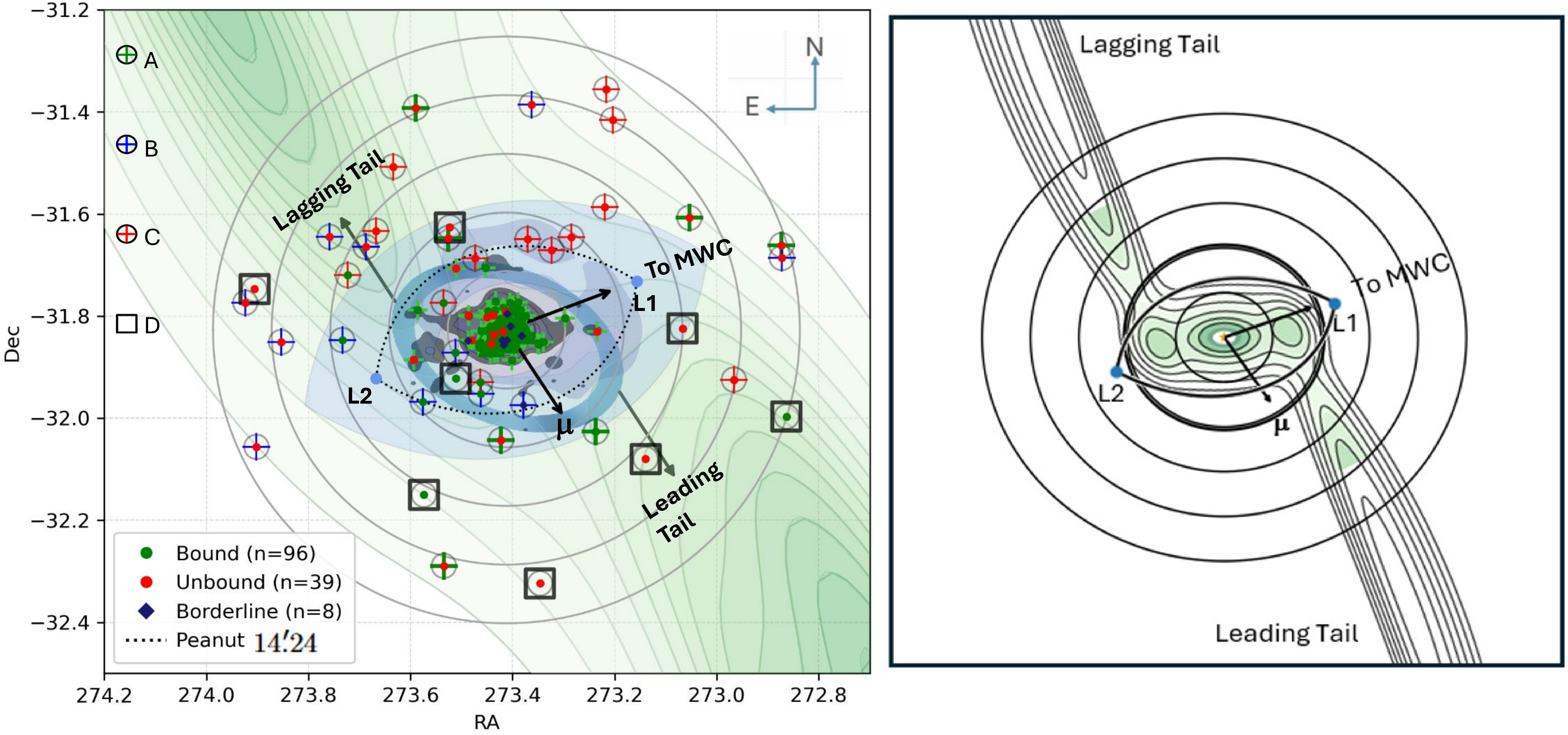




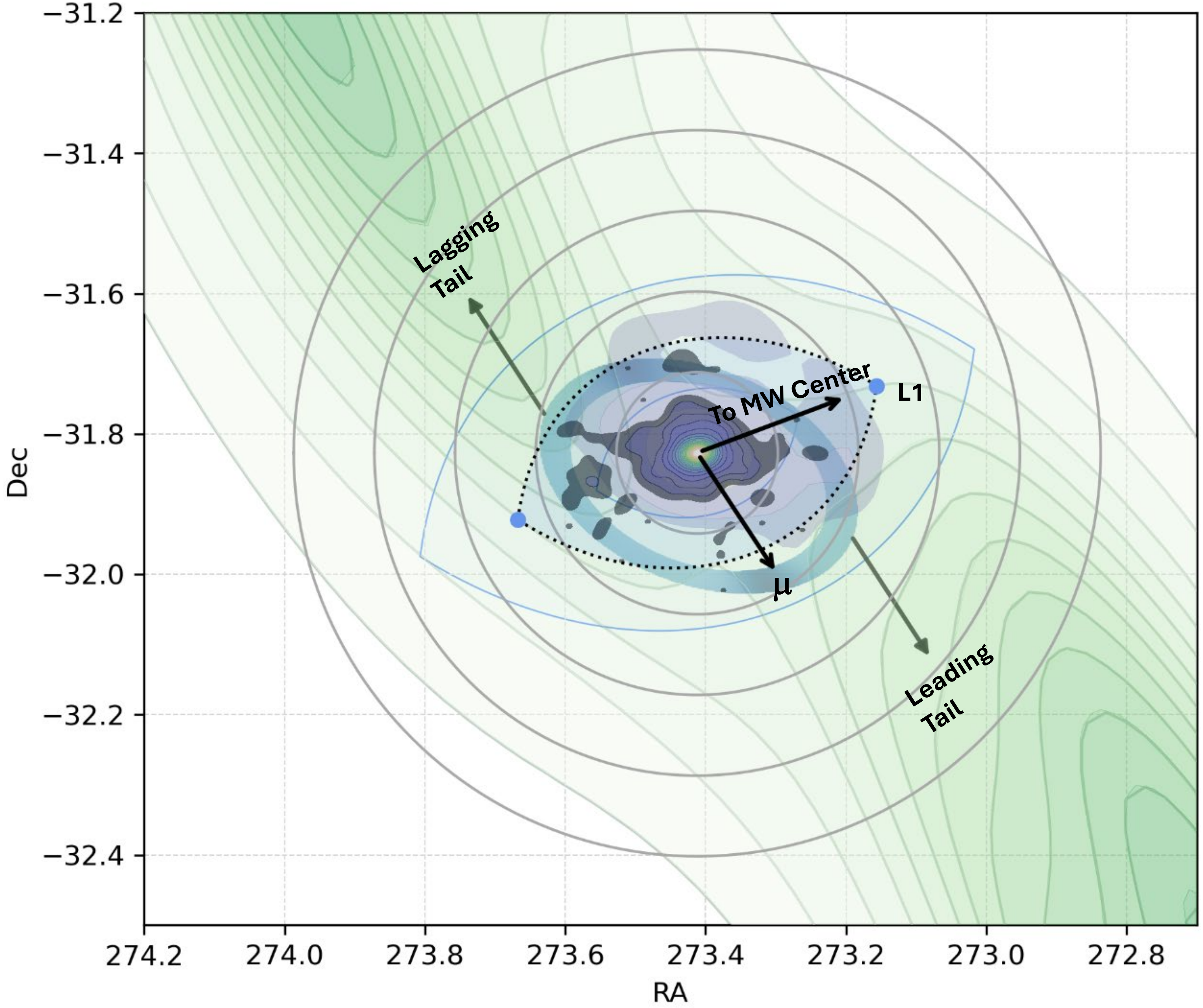
NGC 6569's Orbit over
last 2 Gyr
Fardal Stream
Chen Stream
Sun
MWC
Y (kpc)
X (kpc)
Chen-RV (km s$^{-1}$)

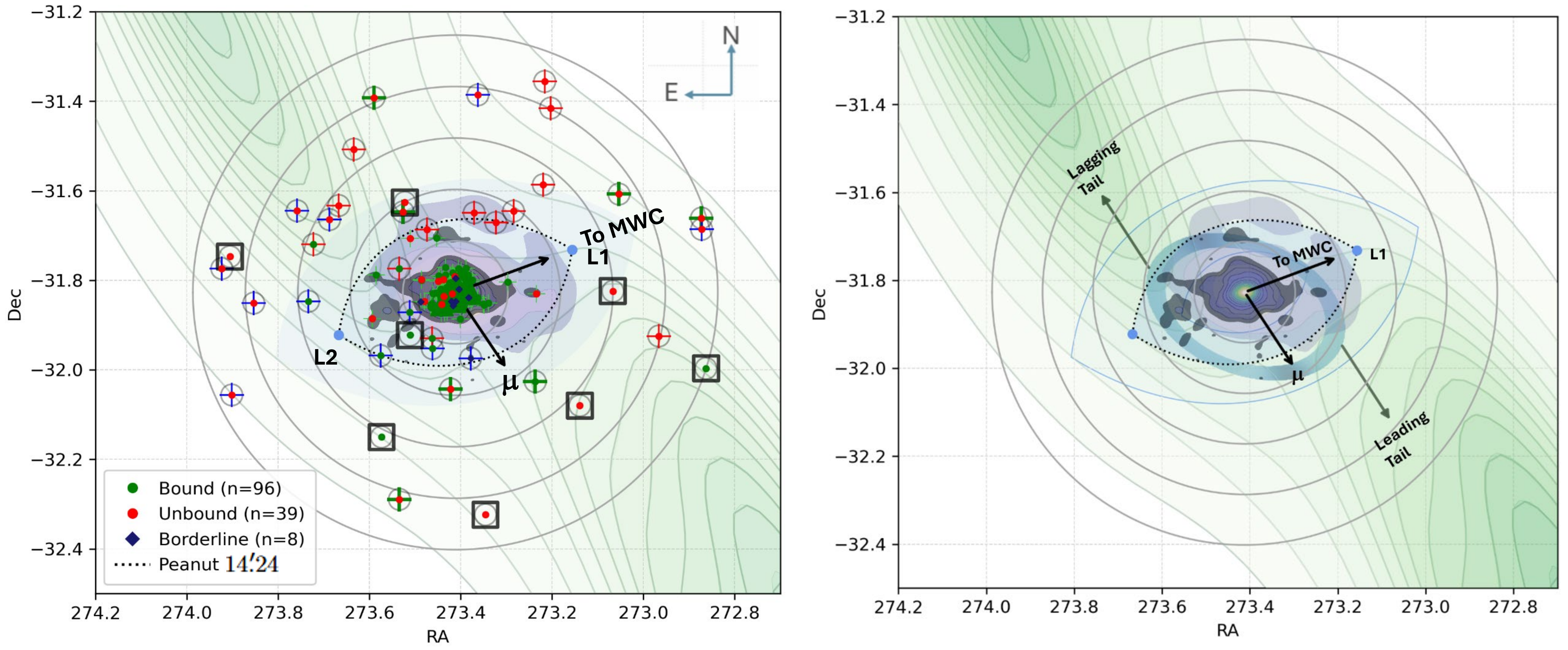
Orbit over past 0.5 Gyr
AAT FOV + PM
Fardal stream (n=344)
Chen stream (n=208)
Johnson+ members (n=100)
rotating bar (de-rotated)
Y [kpc]
X [kpc]
Fraction per bin
RV (km s$^{-1}$)

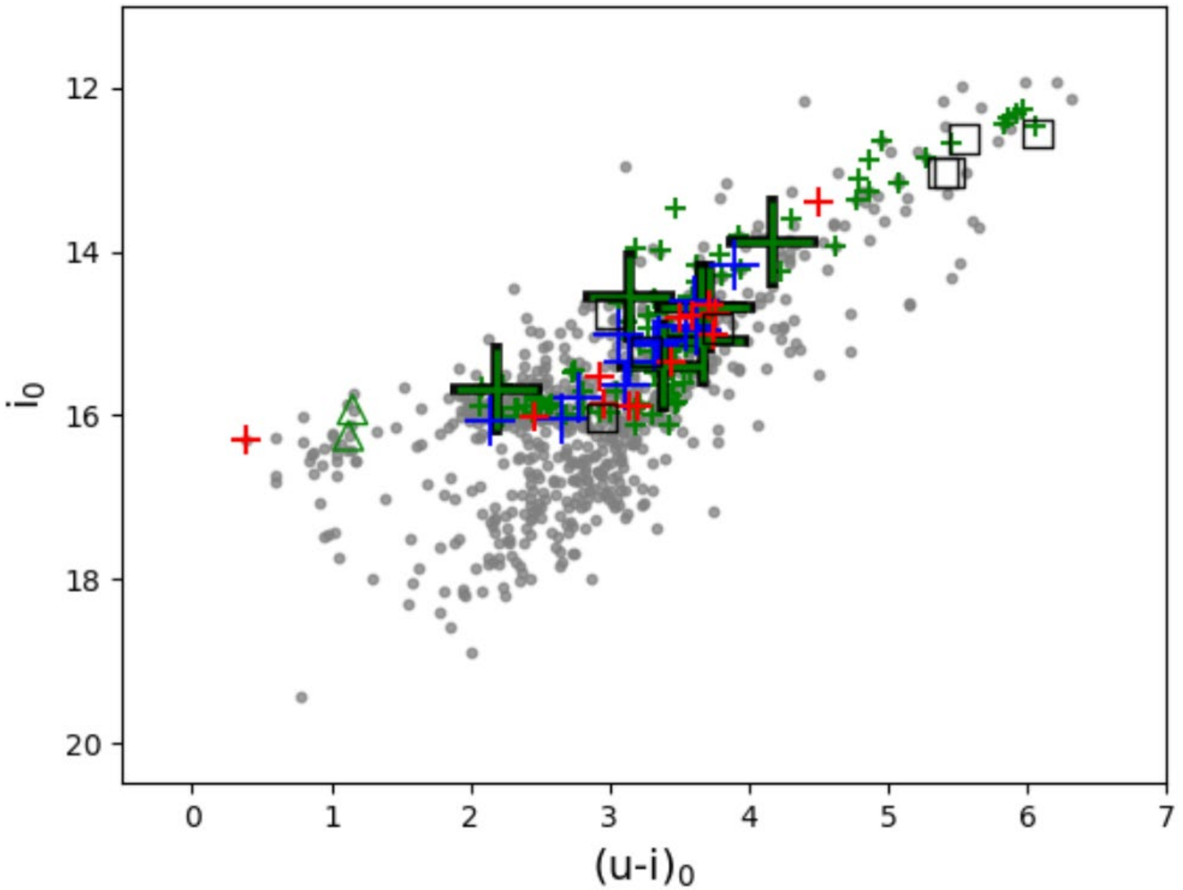


A
B
C
D
N
E
Lagging Tail
To MWC
L1
L2
μ
Leading Tail
Bound (n=96)
Unbound (n=39)
Borderline (n=8)
Peanut 14′24
Dec
RA
−31.2
−31.4
−31.6
−31.8
−32.0
−32.2
−32.4
274.2
274.0
273.8
273.6
273.4
273.2
273.0
272.8
Lagging Tail
To MWC
L1
L2
μ
Leading Tail



−31.2
−31.4
−31.6
−31.8
−32.0
−32.2
−32.4
Dec
274.2
274.0
273.8
273.6
273.4
273.2
273.0
272.8
RA
Lagging Tail
Leading Tail
To MW Center
L1
μ


N
E
To MWC
L1
L2
μ
Bound (n=96)
Unbound (n=39)
Borderline (n=8)
Peanut 14.′24
Dec
RA
Lagging Tail
Leading Tail
To MWC
L1
μ
Dec
RA




Deredded i vs. (u-i) BDBS CMD is shown for objects ASteCA found to be cluster members (gray filled circles). The Objects A-D are shown, from Table 4. Grades:

A: Large green pluses with a black border.

B: Smaller blue pluses.

C: Small red pluses.

D: Open black squares.

The 98/100 of the J18 sample are shown as small green pluses.